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Received: 31 March 2026

Accepted: 2 August 2026

Published online: 09 August 2026

Cite this article as: Vats V., Kim H., Linares E. *et al.* A survey of deep multivariate time-series models with an empirical reproducibility audit. *Artif Intell Rev* (2026). <https://doi.org/10.1007/s10462-026-11674-8>

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A Survey of Deep Multivariate Time-Series Models with an Empirical Reproducibility Audit

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Abstract

Multivariate time series (MTS) analysis is increasingly important for extracting insights from complex, interdependent temporal data in domains such as healthcare, finance, and industrial monitoring. Recent advances in deep learning have significantly improved MTS modeling; yet, the rapidly expanding literature remains fragmented across tasks, architectures, and evaluation practices. This survey concentrates on deep learning-centric approaches in MTS research across seven key tasks: classification, clustering, forecasting, anomaly detection, imputation, representation learning, and transfer learning. To organize the field, we introduce a multi-dimensional taxonomy that categorizes methods along three complementary axes: (i) modeling and learning paradigms (e.g., Transformers, self-supervised contrastive learning, generative models, and emerging LLM-based approaches), (ii) downstream analytical objectives, and (iii) deployment-oriented capabilities such as scalability, robustness to irregular sampling and distribution shifts, interpretability, and transferability. Beyond conceptual synthesis, we conduct a targeted reproducibility audit by re-evaluating representative models using public codebases under controlled experimental conditions. Our findings show that while broad performance trends are often reproducible, exact numerical results can vary across models, datasets, and protocols, particularly when preprocessing or seed-level details are underspecified. Lastly, we identify key challenges, including high dimensionality, sparsity, non-stationarity, and concept drift, and outline seven research directions toward more scalable, reliable, and interpretable MTS systems.

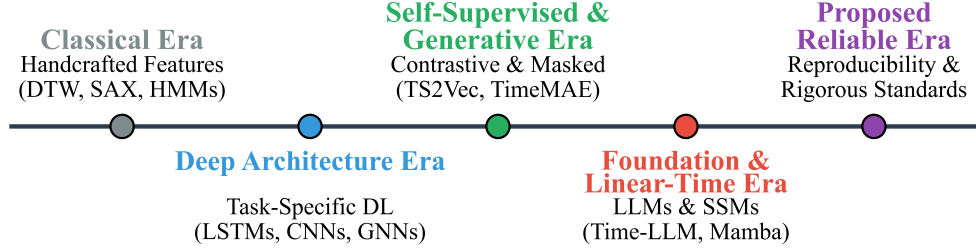


Fig. 1 MTS research lineage: from handcrafted methods to foundation models and reproducibility

Keywords: Multivariate Time Series, Representation Learning, Clustering, Classification, Forecasting, Anomaly Detection, Imputation, Transfer Learning, Downstream Tasks

1 Introduction

Modern systems routinely record multiple variables over time, which produces Multivariate Time Series (MTS) data. Unlike univariate sequences, MTS data encode both temporal dynamics and cross-variable interactions, but they also introduce challenges such as high-dimensional dependencies, missing observations, non-stationarity, and heterogeneous variable relationships (Aghabozorgi et al. 2015; Ismail Fawaz et al. 2019; Lim et al. 2021; Wang et al. 2025, 2024a). From this perspective, MTS analysis is fundamentally a representation learning problem: the goal is to extract latent structures that capture how variables evolve individually and jointly over time.

As illustrated in Fig. 1, MTS research has evolved from classical similarity measures and handcrafted representations to modern deep learning architectures. Early methods relied on lock-step distances, elastic alignment measures such as Dynamic Time Warping, symbolic representations such as SAX and SFA, Shapelets, Matrix Profiles, probabilistic models such as Hidden Markov Models, and ensemble methods such as HIVE-COTE (Deza and Deza 2009; Sakoe and Chiba 2003; Vlachos et al. 2002; Chen and Ng 2004; Stefan et al. 2012; Lin et al. 2003; Schäfer and Höggqvist 2012; Ye and Keogh 2009; Yeh et al. 2016; Rabiner 2002; Lines et al. 2016; Paparrizos et al. 2020). Although effective in many settings, these approaches often depend on handcrafted features, costly similarity computations, or carefully designed symbolic/shape-based representations, which can limit scalability in large-scale and streaming environments.

Deep learning has transformed MTS analysis by enabling representations to be learned directly from raw sequences. Recurrent and convolutional models capture temporal patterns, Graph Neural Networks model cross-variable dependencies, and Transformer-based architectures improve long-range sequence modeling (Hochreiter and Schmidhuber 1997; Cho et al. 2014; Ismail Fawaz et al. 2020; Wu et al. 2019, 2020; Zhang and Yan 2023; Zhou et al. 2022; Nie et al. 2022). More recently, self-supervised

learning and foundation models have expanded the field toward large-scale pretraining and transferable time-series reasoning (Yue et al. 2022; Jin et al. 2023a; Jia et al. 2024).

Despite this rapid progress, the existing deep MTS literature remains *fragmented*. Existing surveys often focus on individual tasks, such as forecasting or anomaly detection, or on specific model families, such as generative models (Benidis et al. 2022; Zamanzadeh Darban et al. 2024; Brophy et al. 2023). As a result, the field lacks a unified view that connects modeling paradigms, analytical tasks, and design capabilities across the broader MTS landscape. This fragmentation makes it difficult to systematically compare methods, identify shared design principles, and understand how modern architectures generalize across different tasks and deployment scenarios.

A second major challenge is *empirical reproducibility*. Reported performance gains often depend on inconsistent datasets, undocumented preprocessing procedures, incomplete hyperparameter specifications, random seeds, and varying evaluation protocols. Consequently, reproducing published results or fairly comparing models remains difficult. Although reproducibility has become a central concern in machine learning, systematic studies examining the reproducibility of deep MTS models remain limited.

These gaps motivate the central scientific and technical question addressed in this survey: *How can the fragmented landscape of deep multivariate time-series analysis be systematically organized into a coherent framework, and to what extent are the reported performance gains of modern deep MTS architectures empirically reproducible under controlled experimental conditions?*

To answer this question, this survey makes two tightly connected contributions. First, we introduce a *three-axis framework* that organizes deep MTS methods by modeling paradigm, analytical task, and design capability. This taxonomy captures the many-to-many structure of modern MTS research: a single architecture may support multiple tasks, while similar model families may differ substantially in scalability, robustness, interpretability, and deployment objectives. Second, we conduct a *targeted reproducibility audit* of representative modern architectures, including Transformer-based, state-space, diffusion, graph, and contrastive models. Using multi-seed experiments and a strict decision rule, we show that implementation details, hyperparameter choices, random seeds, and evaluation protocols can substantially influence reported performance. These results indicate that single-run evaluations often overestimate model stability and highlight the need for routine variance reporting.

The main contributions of this survey are:

- **A Unified Conceptual Framework for Deep MTS Analysis.** We organize deep MTS methods along three complementary dimensions: modeling paradigm, analytical task, and design capability, including scalability, robustness, and interpretability. This taxonomy reveals structural connections across previously isolated research areas.
- **Cross-Task Architectural Insights.** We analyze the architectural mechanisms that support effective MTS modeling, including strategies for capturing long-range temporal dependencies, cross-variable interactions, and robust representations across diverse downstream tasks.
- **Large-Scale Reproducibility Audit.** We conduct a targeted reproducibility audit of representative deep MTS models spanning multiple paradigms and analytical tasks.

Rather than serving as an exhaustive benchmark, this audit reveals how preprocessing choices, hyperparameter specifications, random seeds, evaluation protocols, and implementation details shape empirical reproducibility.

- **Benchmarking Guidelines.** Based on our analysis, we synthesize best practices for dataset preparation, experimental design, and performance reporting to improve transparency, reproducibility, and comparability across MTS studies.
- **Open Challenges and Research Directions.** We highlight key open problems, including scalable modeling of high-dimensional sensor networks, robustness under distribution shifts and missing data, and the integration of foundation models for time-series reasoning.

Overall, this survey does not aim to provide a merely exhaustive catalog of models. Instead, it connects modeling paradigms, downstream tasks, and design capabilities into a unified perspective on deep MTS analysis. By combining conceptual synthesis with empirical verification, the survey provides researchers with a structured basis for evaluating existing methods and developing more reliable MTS architectures.

The rest of this article is organized as follows. Section 2.1 introduces the fundamental terminology of multivariate time-series analysis. Section 2.2 reviews existing surveys and identifies their limitations. Section 3 presents our unified taxonomy of deep MTS methods. Section 4 reports the results of our reproducibility study. Section 5 discusses major challenges in MTS research. Section 6 identifies key research gaps and proposes future directions; Section 7 outlines the limitations of this survey; and Section 8 concludes the paper.

2 Survey Landscape and Research Gaps

Before surveying the methodological landscape, we formalize the data structures underlying multivariate time-series analysis and then review the existing survey literature to identify the research gaps this survey addresses.

2.1 Formalization of Time Series Data

Time-series data consists of ordered observations collected at successive time points, where the temporal ordering carries critical information about system dynamics. Unlike static datasets, time series exhibit temporal dependencies that must be explicitly modeled. This survey focuses on *multivariate time series* (MTS), where multiple variables evolve jointly over time, introducing additional challenges due to cross-variable interactions.

Based on the dimensionality of observations at each timestamp, time series can be categorized as follows.

Definition 2.1 (Univariate Time Series). A sequence of observations of a variable recorded over time, represented as

$$X = (x_0, x_1, x_2, \dots, x_t),$$

where x_i denotes the observation at time step i .

Definition 2.2 (Multivariate Time Series). A collection of multiple variables observed simultaneously over time:

$$X = [X_1, X_2, \dots, X_p],$$

where each X_i is a univariate time series and p denotes the number of variables.

Definition 2.3 (Multiple Multivariate Time Series). A dataset consisting of multiple MTS instances:

$$\mathcal{X} = \{X^{(1)}, X^{(2)}, \dots, X^{(n)}\},$$

where each $X^{(j)}$ represents an independent multivariate time series instance, and n denotes the total number of MTS instances in the dataset.

2.2 Existing Surveys and Research Gaps

The rapid expansion of deep learning research for multivariate time-series analysis has led to a growing number of survey articles. However, as illustrated by the task coverage in Table 1, existing surveys typically focus on narrow aspects of the field. Although these works provide valuable insights, the overall survey landscape remains *fragmented*. To provide context for this growing body of literature, Table 2 provides a detailed summary of major recent surveys, outlining the methods they cover, key features, and inherent limitations.

In the following subsections, we categorize prior surveys into three groups: task-specific surveys, model-centric surveys, and emerging surveys on large language models (LLMs) and multimodal approaches.

2.2.1 Task-Specific Surveys

A significant portion of existing surveys is dedicated to individual downstream tasks. Forecasting surveys analyze deep learning architectures, particularly Transformer-based models, primarily in the forecasting setting (Benidis et al. 2022; Wen et al. 2022), particularly Transformer-based models, but primarily in the forecasting setting. Similarly, (Zamanzadeh Darban et al. 2024) covers anomaly detection methods and (Wang et al. 2024a) reviews imputation techniques.

Other surveys investigate self-supervised representation learning for time series. For example, (Zhang et al. 2024a) and (Trirat et al. 2024) examine contrastive learning approaches, such as TS2Vec (Yue et al. 2022). However, these works primarily analyze representation learning as an isolated objective and rarely examine how learned representations transfer across downstream tasks.

Overall, task-specific surveys provide detailed analyses within individual problem settings but do *not* offer a unified perspective across different MTS tasks.

2.2.2 Model-Centric Surveys

Another group of surveys organizes the literature around specific model families. Transformer-based architectures for time-series forecasting have been explored with

Table 1 Coverage of downstream tasks in recent MTS survey papers.

Downstream Task	(Wang et al. 2024a)	(Benidis et al. 2022)	(Zamanzadeh Darban et al. 2024)	(Brophy et al. 2023)	(Paparrizos et al. 2024)	(Liu et al. 2025)	(Zhang et al. 2024b)	(Zhang et al. 2024a)	(Wen et al. 2022)	(Jin et al. 2024)	(Jin et al. 2023b)	(Tiriat et al. 2024)	(Shi et al. 2025)	(Cini et al. 2025)	Ours
Forecasting	✗	✓	✗	✗	✗	✓	✓	✓	✓	✓	✓	✗	✓	✓	✓
Classification	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✗	✗	✗	✓
Anomaly Detection	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✗	✓	✗	✓
Clustering	✗	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓
Imputation	✓	✗	✗	✓	✗	✓	✗	✓	✗	✓	✗	✗	✗	✗	✓
Representation Learning	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓	✓	✗	✗	✓
Transfer Learning	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓

emphasis on mechanisms for modeling long-range dependencies (Wen et al. 2022). Similarly, (Jin et al. 2024) and (Cini et al. 2025) investigate Graph Neural Networks (GNNs) for spatio-temporal modeling, particularly when explicit graph structures are available.

Generative modeling approaches, including TimeGAN, have also been examined for time-series data generation and imputation (Brophy et al. 2023). While these surveys provide detailed architectural analyses, their scope is restricted to specific model classes and does not extend to broader methodological comparisons across tasks.

2.2.3 Emerging Surveys on LLMs and Multimodal Learning

More recently, several surveys have explored integrating large language models with time-series analysis. Surveys like (Liu et al. 2025), (Zhang et al. 2024b), and (Shi et al. 2025) focused on approaches such as temporal tokenization and prompt-based adaptation of LLMs for time-series forecasting.

That said, these surveys remain largely conceptual and often lack extensive empirical evaluation. Moreover, they rarely connect LLM-based methods with traditional multivariate time-series tasks such as anomaly detection, clustering, or imputation.

2.2.4 Key Limitations of Existing Surveys

Despite the breadth of existing surveys, several important gaps still exist, such as:

Table 2 Summary of the major surveys concerning MTS analysis

Paper	Methods Covered	Key Features	Limitations
(Paparrizos et al. 2024)	Distance-, partitional-, hierarchical-, distribution-, subsequence-, shapelet-based; representation learning; comparative/generative; foundation/LLM-based	Comprehensive taxonomy from classic to modern deep/LLM-based TS clustering; highlights efficiency, interpretability, embedding richness	Distance-sensitive; costly elastic measures; k-means initialization sensitivity; fuzzy methods add compute cost; hierarchical slow; model/density assumptions; feature/encoding info loss; subsequence costly; representation learning augmentation-dependent; foundation models immature
(Liu et al. 2025)	Prompt design; cross-modal alignment; fusion strategies	Integrates LLMs with TS via prompting, alignment, fusion; multimodal representation taxonomy	Prompt/token info loss; retrieval/contrastive heavy; distillation complexity; simple fusion misses dependencies
(Zhang et al. 2024b)	Direct prompting; quantization; alignment; vision-as-bridge; tool integration	Adapts LLMs to TS via prompting, token conversion, visual encoding, tool use	Numeric format mismatch; discretization errors; alignment complexity; visual conversion overhead; tool dependency
(Zamanzadeh Darban et al. 2024)	Forecasting-, reconstruction-, hybrid-based	Forecasting uses error; reconstruction with AEs/VAEs/GANs; hybrid boosts robustness	Forecasting misses subtle anomalies; reconstruction overfits; hybrid costly
(Zhang et al. 2024a)	Generative-, contrastive-, adversarial-based SSL	SSL taxonomy with pretext tasks (forecasting, reconstruction, alignment, GANs); focuses on robustness, domain adaptation	Generative overfits; contrastive sensitive to sampling; adversarial unstable; high compute
(Benidis et al. 2022)	RNN-, CNN-, Seq2Seq-, hybrid-, probabilistic-, GAN-based	Core DL forecasting models; temporal dependency, multi-horizon, probabilistic outputs, interpretability	RNNs long-dependency issues; CNNs limited context; Seq2Seq tuning complex; hybrids heavy; GANs unstable
(Wen et al. 2022)	Transformer variants; positional encoding; task-specific modifications	Improved attention, hybrid architectures, encoding, domain adaptation	Quadratic cost; encoding misalignments; tuning heavy; dataset-specific drop
(Jin et al. 2024)	GNN-based forecasting/classification/anomaly/imputation	Spatial-temporal modeling via static/dynamic graphs	Graph design hard; scaling issues; noisy edges
(Jin et al. 2023b)	LLMs vs PFMs for TS/ST data	Unified taxonomy; datasets, models, tools; future roadmap	Broad scope; less depth per method; limited empirical validation
(Wang et al. 2024a)	Predictive vs generative; RNN-, CNN-, GNN-, Transformer-, PFM-, LLM-based	Taxonomy by uncertainty: deterministic $\rightarrow$ aleatoric (probabilistic/generative) $\rightarrow$ epistemic (Bayesian/ensembles) + mechanism/shift-aware	Predictive no uncertainty; generative costly; RNNs long-dep.; CNNs miss long-range; GNNs scaling; Transformers heavy; PFMs/LLMs resource-hungry
(Trirat et al. 2024)	Architecture enhancements; learning objectives; data strategies	Universal representation learning via model design, task-agnostic losses, augmentation	Complex tuning; labels for supervised; SSL augmentation-sensitive; augmentation artifacts
(Shi et al. 2025)	Input-stage; optimization; deployment	LLM adaptation via tokenization, fine-tuning, efficient deployment	Token/prompt sensitive; fine-tuning costly; compression accuracy loss; data scarcity
(Brophy et al. 2023)	Discrete-/continuous-GANs; privacy/evaluation-aware; task-specific GANs	GAN adaptations for discrete/continuous TS, privacy, anomaly detection, classification	Discrete gradient issues; continuous long-dep.; privacy may lower fidelity; task-specific poor generalization
(Cini et al. 2025)	STGNNs; message-passing blocks; global-local modeling	Graph-based forecasting with spatial-temporal dependencies; balance global/local	Graph design complex; scaling issues; tuning heavy; limited benchmarking

Task fragmentation. Most surveys focus on individual tasks (e.g., forecasting or anomaly detection) rather than analyzing the relationships among multiple MTS problems.

Architecture-centric perspectives. Model-focused surveys emphasize particular architectures (e.g., Transformers or GNNs) but do *not* systematically compare them across tasks.

Limited empirical verification. Existing surveys typically summarize reported results from original papers *without* conducting independent experimental validation.

Incomplete coverage of emerging paradigms. Recent developments such as transfer learning, multimodal modeling, and large-scale representation learning remain underrepresented in the survey literature.

2.2.5 Position of This Survey

This survey addresses the above limitations by providing a unified and empirically grounded synthesis of deep learning methods for multivariate time-series analysis.

Specifically, our work differs from prior surveys in three key aspects:

- **Unified cross-task perspective.** We jointly analyze seven major MTS tasks: forecasting, anomaly detection, classification, clustering, imputation, representation learning, and transfer learning within a single framework.
- **Multi-dimensional taxonomy.** We organize existing methods along three complementary dimensions: learning paradigms, model architectures, and downstream tasks. This framework reveals shared design principles across previously disconnected research areas.
- **Reproducibility-oriented evaluation.** Beyond summarizing prior results, we conduct a targeted reproducibility study of representative deep learning models under standardized experimental conditions.

By integrating these perspectives, our survey provides a unified view of the rapidly evolving MTS research landscape and highlights directions for future work.

3 A Multi-Dimensional Taxonomy of Deep MTS Methods

To organize the rapidly expanding literature on multivariate time-series (MTS) analysis, we propose a multi-dimensional taxonomy that characterizes methods along three complementary axes: (Axis 1) the modeling and learning paradigm, which defines *how* representations are learned; (Axis 2) the analytical task, which defines *what* the model predicts or optimizes; and (Axis 3) the design capability, which explains *why* a method is engineered in a particular way under practical constraints such as scalability, robustness, or interpretability.

Figure 2 illustrates our multi-dimensional taxonomy of MTS research involving deep learning. This taxonomy reflects the many-to-many structure of modern MTS research. A single backbone can support multiple downstream tasks, and the same architectural family can be optimized for different deployment objectives. Rather

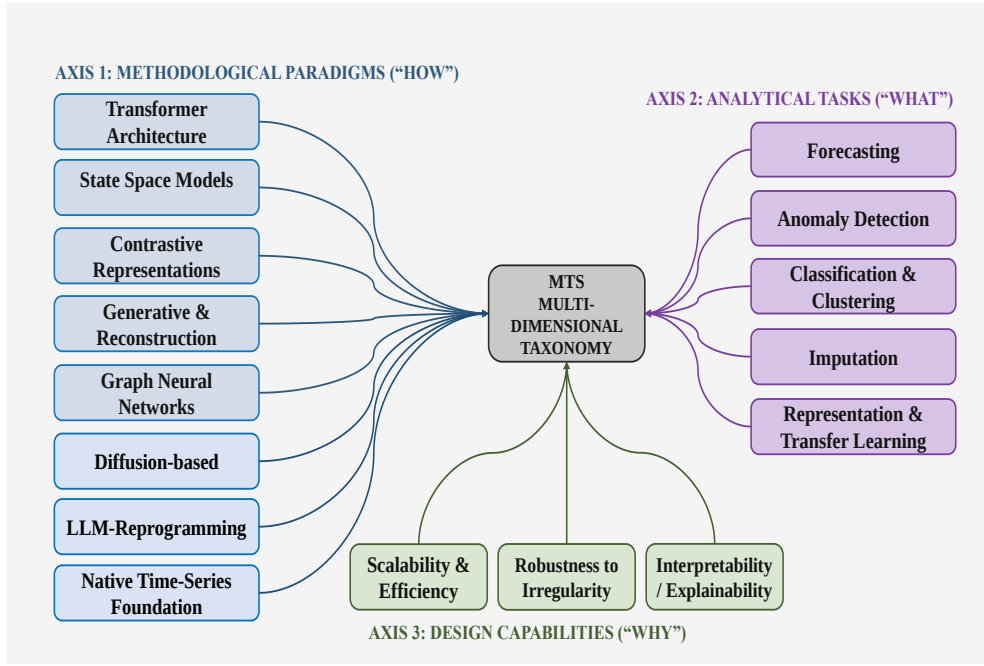


Fig. 2 Overview of the proposed three-axis taxonomy for multivariate time-series research. Methods are categorized by modeling paradigm (Axis 1), downstream analytical tasks (Axis 2), and deployment-oriented design capabilities (Axis 3).

than forcing methods into mutually exclusive categories, our taxonomy exposes the relationships among paradigms, tasks, and design choices.

3.1 How to Read the Taxonomy

We treat taxonomy assignment as *multi-label* rather than mutually exclusive. Each method is associated with: (i) one primary Axis-1 tag indicating its dominant modeling paradigm, (ii) one or more Axis-2 tags denoting downstream tasks, and (iii) one or more Axis-3 tags capturing the capabilities explicitly targeted by the design.

This formulation better reflects modern MTS models, which are often hybrid. For example, a method may combine masked reconstruction with contrastive learning or integrate Transformer backbones with graph priors while still serving multiple downstream tasks.

3.2 Axis 1: Modeling Paradigms (The “How”)

Axis 1 classifies methods by the primary modeling paradigm used to capture temporal dynamics and cross-variable dependencies. Recent deep MTS research is organized around several recurring paradigms: *Transformer-based models*, *graph neural networks*,

state-space models, self-supervised contrastive learning, generative and reconstruction-based models, diffusion models, LLM-reprogramming methods, and native time-series foundation models.

We separate LLM-reprogramming from native time-series foundation models to preserve conceptual precision. The former adapts pretrained text-language models by mapping numerical sequences into token-like, patch-based, or prompt-based inputs, whereas the latter learns temporal representations through direct pretraining on large-scale time-series corpora. This distinction clarifies how each paradigm acquires MTS representations, even as hybrid designs continue to emerge. Table 3 summarizes the strengths, limitations, and typical applications of each paradigm.

3.2.1 Transformer-Based Models

Transformer architectures have become a dominant backbone for MTS analysis because self-attention can model long-range temporal dependencies more effectively than recurrent architectures. In MTS settings, attention may be applied along the temporal axis, across variables, or jointly across both dimensions, depending on whether the design prioritizes long-range forecasting, cross-channel interaction, or both.

A central challenge is the quadratic complexity of attention with respect to sequence length. This has motivated a series of efficiency-oriented designs. PatchTST (Nie et al. 2022), for instance, reduces token length by segmenting each series into patches, substantially lowering attention costs while improving robustness to point-wise noise. In contrast, Crossformer (Zhang and Yan 2023) emphasizes explicit cross-dimension modeling through two-stage attention, arguing that inter-variable dependencies often carry the primary signal in MTS data. More recent hybrids seek to balance these two perspectives by combining temporal patching with channel interaction modules.

Linear models like DLinear (Zeng et al. 2023) have recently challenged these complex mechanisms. They proved that simple baselines are often highly competitive. In response, iTransformer (Liu et al. 2023b) inverted the Transformer paradigm entirely. It embeds full variables as independent tokens to better capture multivariate correlations.

Taken together, Transformer-based MTS models differ significantly in how they balance temporal resolution, cross-channel modeling, and computational efficiency. This design space has become especially influential in forecasting, imputation, and representation learning.

3.2.2 Graph Neural Networks

Unlike architectures that treat variables independently or uniformly, GNNs explicitly model cross-channel dependencies as graph-structured relationships. Through message-passing mechanisms, GNN layers propagate information between connected nodes, enabling the modeling of spatial and relational interactions among variables. This design makes GNNs particularly well-suited for structured domains, including sensor networks, transportation systems, and industrial monitoring environments. That said, constructing and updating dynamic graphs can incur significant computational overhead, often scaling quadratically with the number of variables ($\mathcal{O}(L \cdot D^2)$). To improve efficiency, recent architectures such as Lite-STGNN (Moges and Moodley

Table 3 Analysis of Methodological Paradigms Mapped to Analytical Tasks and Design Capabilities

Methodologi Paradigm	Core Insight	Key Strengths	Main Limitations	Analytical Tasks	Design Capabilities
<i>Transformer- Based Models</i>	Self-attention applied temporally, across channels, or jointly.	Dominant for capturing complex, long-range dependencies.	Quadratic complexity ($\mathcal{O}(L^2)$) without patching mechanisms.	Forecasting, Imputation, Representation Learning	Long-horizon modeling, variable-length sequence handling.
<i>Graph Neural Networks (GNNs)</i>	Message-passing layers over explicit topological structures.	Unparalleled structural interpretability for spatially grounded data.	Quadratic cost for dynamic graph construction ($\mathcal{O}(L \cdot D^2)$).	Spatiotemporal Forecasting, Anomaly Detection	Explicit relational modeling, robustness to non-stationarity.
<i>State-Space Models (SSMs)</i>	Continuous-time sequence processing via state-space matrices.	Highly efficient for ultra-long sequences with linear-time processing.	Injecting cross-channel structure often compromises linear efficiency.	Long-horizon Forecasting, Classification	Strict linear scalability ($\mathcal{O}(L)$), extreme sequence efficiency.
<i>Self- Supervised Contrastive Learning</i>	Positive/negative pair contrast via shapelets or frequency domains.	Excellent for transferable representations under strict label scarcity.	Downstream performance relies heavily on the quality of positive/negative pairs.	Representation Learning, Classification, Clustering	Domain adaptation, robustness to spectral corruption.
<i>Generative & Recon- struction</i>	Masked autoencoding and partial observation recovery.	Strong inductive bias for recovering corrupted or missing data.	Less effective for complex probabilistic generation than diffusion.	Imputation, Anomaly Detection	Robustness to partial observations and missing values.
<i>Diffusion Models</i>	Iterative forward noise injection and learned reverse denoising.	State-of-the-art probabilistic forecasting and uncertainty quantification.	Severe computational overhead during iterative inference limits deployment.	Probabilistic Forecasting, Imputation	Uncertainty estimation, handling irregular or NMAR data.
<i>LLM- Reprogram. / Adaptation</i>	Reuses pretrained text LLMs by converting time series into tokens, patches, or prompts.	Strong few-shot and cross-domain adaptation.	Sensitive to tokenization, prompting, normalization, and high compute cost.	Forecasting, anomaly detection, classification, imputation	Cross-modal adaptation, transferability.
<i>Native Time-Series Foundation Models</i>	Pretrains large backbones directly on time-series corpora.	Learns time-series-native representations for zero/few-shot transfer.	High pretraining cost, possible data leakage, and limited corpus transparency.	Forecasting, imputation, representation learning, transfer learning	Unified task handling, cross-domain transfer.

2025) employ strategies including adjacency factorization and graph sparsification to reduce computational complexity while maintaining predictive performance.

Modern applications increasingly require modeling higher-order and evolving relationships beyond static pairwise connections. To address this need, recent work has incorporated directed hypergraph structures that capture group-wise interactions among variables, enabling the modeling of phenomena such as simultaneous sensor drift (Ha and Kim 2025). Other architectures, such as FC-STGNN (Wang et al. 2024b), learn fully

connected correlations across variables and temporal dimensions simultaneously. Additionally, memory-augmented Mixture-of-Experts (MoE) routing mechanisms (Huang et al. 2025) have been introduced to dynamically allocate specialized graph-processing modules for heterogeneous relational patterns. These developments aim to improve the scalability of dynamic graph structures for high-dimensional multivariate time-series data, while improving flexibility and expressiveness. However, there remain open challenges.

3.2.3 State-Space Models

State-space models (SSMs), especially selective SSMs such as Mamba, have recently emerged as an alternative to Transformers for long-horizon time-series modeling. Their main appeal lies in linear-time sequence processing, which makes them attractive for long sequences where full attention becomes costly.

Recent MTS variants extend SSMs to better capture variable interactions and multi-scale structures. For example, MambaTS (Cai et al. 2024b) introduces variable-aware scanning to impose structure over multivariate dependencies, while models such as TimeMachine (Ahamed and Cheng 2024), CMamba (Zeng et al. 2024), and MambaMixer (Behrouz et al. 2024) enhance channel mixing and cross-variable modeling. These methods suggest that the main challenge for SSMs in MTS is not temporal modeling itself, but how to inject multivariate structure without sacrificing linear scalability.

Overall, SSM-based methods are best understood as efficiency-oriented sequence backbones whose effectiveness depends on how well they reconcile long-range temporal modeling with cross-channel interactions.

3.2.4 Self-Supervised Contrastive Learning

Contrastive learning has become a major paradigm for learning transferable MTS representations from unlabeled data. Rather than optimizing a task-specific supervised objective, these methods learn an embedding space in which semantically similar segments are pulled together, and dissimilar ones are pushed apart.

The main design differences lie in how positive and negative pairs are constructed. TS2Vec (Yue et al. 2022) contrasts multiple views of the same series at both timestamp and contextual levels, while later methods such as TimeCSL (Liang et al. 2024), CaTT (Shamba et al. 2025), and CoST (Woo et al. 2022) refine the notion of similarity through shapelet structure, efficient positive mining, or seasonal-trend disentanglement. Another line of work, including TimesURL (Liu and Chen 2024) and TF-C (Zhang et al. 2024a), extends contrastive consistency into the frequency domain to preserve spectral information.

These methods are particularly important for representation learning, classification, anomaly detection, and transfer learning, where label scarcity and domain shift often limit purely supervised approaches.

3.2.5 Reconstruction-based and Masked Autoencoding Models

Reconstruction-based methods learn MTS representations by recovering masked, corrupted, or partially observed temporal signals. Their core assumption is that successful reconstruction requires the model to encode temporal structure and cross-variable dependencies. Masked autoencoding is a prominent realization of this paradigm: methods such as TimeMAE (Cheng et al. 2023) and Ti-MAE (Li et al. 2023) reconstruct masked patches or segments from incomplete inputs, encouraging the encoder to capture long-range temporal dependencies and inter-variable relationships. These representations can then support forecasting, imputation, anomaly detection, and downstream representation learning.

Compared with contrastive approaches, reconstruction-based methods provide a more direct inductive bias for tasks involving missing-value recovery or corrupted-signal restoration. However, their effectiveness depends heavily on the masking strategy, reconstruction target, and alignment between the pretext task and realistic downstream missingness patterns. This limitation is especially important when missing values are not uniformly random but arise from sensor failures, clinical measurement policies, or other structured mechanisms.

3.2.6 Diffusion-based Models

Diffusion models constitute a distinct generative paradigm for probabilistic forecasting and imputation. Unlike masked reconstruction methods that typically recover missing values in a single forward pass, diffusion-based models learn an iterative denoising process that transforms noise into structured temporal signals. This formulation supports predictive distribution modeling and provides a natural mechanism for uncertainty estimation.

Recent work adapts diffusion processes to better capture sequential structure, since standard formulations often rely on independent Gaussian noise. For example, autoregressive moving diffusion (Gao et al. 2025) incorporates autoregressive and moving-average components to model temporal dependence, while hybrid architectures use Transformer-derived priors to guide the denoising trajectory and capture complex covariate interactions (Li et al. 2024).

Diffusion-based models are also useful for irregularly sampled or partially observed time series, where generative modeling can produce plausible imputations. Physics-informed variants further incorporate domain constraints into the denoising process, improving robustness for scientific and industrial anomaly detection (Soni et al. 2025). Despite these strengths, diffusion models often require multiple inference-time denoising steps, resulting in substantial computational overhead and limiting their practicality in low-latency or real-time streaming settings.

3.2.7 LLM-Reprogramming and Adaptation Methods

LLM-reprogramming and adaptation methods reuse pretrained text-language models for time-series analysis. Their defining characteristic is that the backbone model is originally trained on natural-language corpora rather than on time-series data. To make numerical temporal sequences compatible with such models, these methods transform

time-series values into token-like embeddings, patches, prompts, or textual descriptions. Thus, the resulting model does not learn a time-series foundation from scratch; instead, it attempts to transfer the sequence-modeling and reasoning capabilities of a frozen or partially fine-tuned LLM to temporal data through modality-bridging modules.

Representative examples include Time-LLM (Jin et al. 2023a), GPT4MTS (Jia et al. 2024), and aLLM4TS (Bian et al. 2024). These methods are attractive because they can leverage the broad transferability of large pretrained LLMs and may support few-shot or cross-domain adaptation with limited task-specific training. However, their performance depends critically on how numerical sequences are normalized, tokenized, prompted, and aligned with the language-model input space. This dependence introduces additional sources of preprocessing sensitivity and computational overhead. Furthermore, because the underlying backbone is optimized for text rather than numerical temporal dynamics, temporal structure may be represented only indirectly through the chosen reprogramming or alignment strategy.

3.2.8 Native Time-Series Foundation Models

Native time-series foundation models are conceptually distinct from LLM-reprogramming methods. Their defining characteristic is that the backbone is pretrained directly on large-scale time-series corpora rather than inheriting it from a text-pretrained LLM. These models learn temporal representations natively from numerical sequences and are typically trained using forecasting, masked prediction, reconstruction, or multi-task objectives. The goal is to construct general-purpose time-series backbones that can be transferred across datasets, domains, horizons, and downstream tasks, including forecasting, imputation, classification, and anomaly detection.

Representative examples include Chronos (Ansari et al. 2024), TimesFM (Das et al. 2024), MOIRAI (Woo et al. 2024), UniTS (Gao et al. 2024), and TimeGPT (Garza et al. 2023). Unlike LLM-reprogramming methods, these models do not rely primarily on converting time-series data into a form acceptable to a text-language model. Instead, they scale time-series-specific pretraining itself, thereby aiming to learn reusable temporal patterns across heterogeneous domains. Their main strengths are cross-domain transferability, unified task handling, and zero-shot or few-shot adaptation. At the same time, they introduce new challenges, including high pretraining costs, potential benchmark contamination, limited transparency into pretraining corpora, and unresolved questions about fair evaluation across heterogeneous datasets.

Note that this separation clarifies our taxonomy: LLM-reprogramming methods are categorized by *reuse of text-pretrained LLM backbones through temporal-to-language alignment*, whereas native time-series foundation models are categorized by *direct pretraining on large-scale time-series data*. Although both lines of work pursue transferable temporal modeling, they differ fundamentally in their training data sources, input representations, architectural assumptions, and evaluation criteria.

3.3 Axis 2: Analytical Tasks (The “What”)

Axis 2 organizes methods by downstream analytical objectives. In this survey, we consider seven core MTS tasks: (i) forecasting, (ii) anomaly detection, (iii) classification,

(iv) clustering, (v) imputation, (vi) representation learning, and (vii) transfer learning. These can be viewed more broadly as three macro-families: *predictive tasks* (forecasting, imputation), *discriminative tasks* (anomaly detection, classification, clustering), and *cross-cutting learning settings* (representation learning, transfer learning).

This axis is important because the same backbone often behaves differently across tasks. For example, forecasting emphasizes temporal extrapolation, anomaly detection emphasizes deviation from normal dynamics, and transfer learning emphasizes representation robustness across domains.

3.3.1 Forecasting

Forecasting is the most studied MTS task and remains the primary benchmark for many modern architectures. Its central challenge is to extrapolate future trajectories from historical observations while maintaining stability over long horizons, capturing cross-variable interactions, and adapting to dynamic distributional changes. This makes forecasting a key testbed for evaluating both predictive accuracy and robustness in complex temporal environments.

Recent work reflects four major design choices. First, efficient temporal modeling has become central, motivating patch-based Transformers such as PatchTST (Nie et al. 2022), linear-time alternatives such as Mamba-based models, and lightweight architectures for resource-constrained deployment. Second, the field remains divided between channel-independent and explicit cross-channel modeling strategies, with PatchTST and iTransformer (Liu et al. 2023b) representing two influential but contrasting directions. Third, recent methods increasingly integrate external modalities, such as text or event context, to improve forecasting when numerical signals are semantically sparse. Fourth, biologically inspired spiking or pulse neural network structures have emerged as a promising direction for efficient MTS prediction.

Hybrid spiking-neuron-embedded LSTM models integrate recurrent temporal modeling with event-driven spiking computation for multivariate time-series prediction under concept-drift environments (Zheng et al. 2022). By activating only in response to informative temporal changes, spiking mechanisms offer a promising route to lightweight, energy-efficient forecasting, especially when data streams are non-stationary or computational resources are limited. Beyond general-purpose MTS benchmarks, lightweight prediction is also critical in latency-sensitive application domains such as instant delivery, where service-route and arrival-time estimation must operate under dynamic, large-scale, and resource-constrained conditions (Wen et al. 2024). These studies highlight an emerging direction in MTS forecasting that complements mainstream deep architectures by emphasizing temporal adaptability, computational efficiency, and deployment-oriented design.

3.3.2 Anomaly Detection

Anomaly detection aims to identify points, subsequences, or entire series that deviate from expected behavior. In deep MTS analysis, this task is commonly addressed through reconstruction error, representational discrepancy, forecasting error, or graph-based relational modeling.

Reconstruction-based methods regard anomalies as patterns that cannot be faithfully recovered by models trained on normal data, while methods such as Anomaly Transformer (Xu et al. 2021) exploit discrepancies in temporal association patterns. Hybrid reconstruction-forecasting approaches further capture both local abnormal patterns and temporal deviations; for example, convolutional residual autoencoding combined with ConvLSTM-based forecasting has been used to model multivariate anomalies from complementary perspectives (Chen et al. 2024). Recent studies also incorporate cross-scale structure, graph topology, and more rigorous benchmarking, reflecting a broader shift from isolated anomaly scoring toward more reliable and realistic evaluation settings.

Application-oriented studies further underscore the importance of MTS anomaly detection in security-sensitive and industrial environments. In these settings, privacy preservation, scalable monitoring, and robust identification of abnormal system behavior are central concerns (Anjum et al. 2025; Chen et al. 2024). These developments indicate that anomaly detection is not merely a benchmark task, but a deployment-critical capability for real-world MTS systems.

3.3.3 Classification and Clustering

Classification and clustering emphasize learning discriminative temporal representations rather than predicting future values. These tasks are essential in domains where discovering underlying patterns is more critical than forecasting, particularly when labeled data is scarce or decision interpretability is required.

Recent advancements in this space generally follow two major directions. First, *shape-based modeling* enhances interpretability by grounding decisions in characteristic subsequences. Second, *self-supervised representation learning* specifically through contrastive and hybrid generative methods learns robust latent spaces that naturally separate complex sequences, providing powerful embeddings for both unsupervised clustering and supervised classification.

3.3.4 Imputation

Imputation estimates missing values in partially observed MTS data and is critical in real-world sensor, healthcare, and industrial settings. Modern approaches increasingly treat imputation as a structured generation problem rather than a purely local interpolation task.

Masked autoencoding provides a natural mechanism for time-series imputation by reconstructing missing observations from the global temporal context learned during pretraining. By randomly masking portions of the input sequence and training the model to recover them, these architectures learn representations that capture long-range temporal dependencies. Diffusion-based models, such as CSDI (Tashiro et al. 2021), extend this paradigm by explicitly modeling uncertainty through conditional generative processes that sample plausible values for missing observations.

Building on this probabilistic framework, recent architectures like Diffusion Transformers (Ye and Chen 2025) further integrate diffusion modeling with Transformer-based sequence representations, enabling more expressive modeling of temporal dependencies while improving uncertainty estimation. Moreover, emerging multimodal approaches

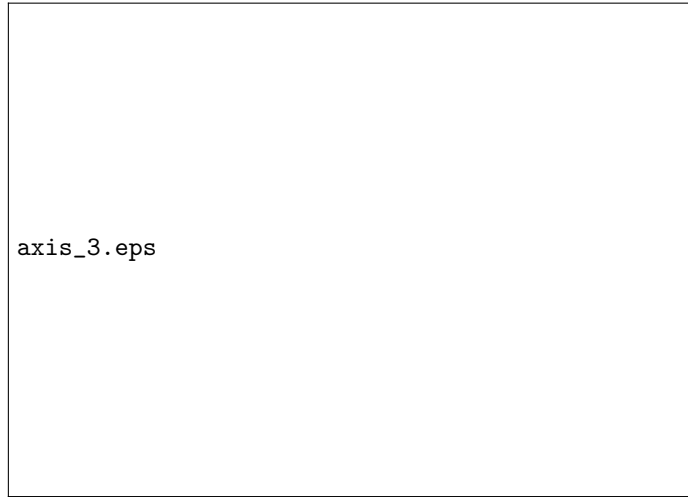


Fig. 3 MTS Paradigm Trade-offs: This radar chart illustrates the design trade-offs among model families across Axis 3 capabilities, such as SSMs’ focus on scalability versus the superior transferability and generalization of LLM-based approaches.

incorporate auxiliary signals, such as metadata, spatial context, or exogenous variables, to improve imputation accuracy when missingness is correlated with broader system dynamics.

3.3.5 Representation and Transfer Learning

Representation learning and transfer learning cut across all MTS tasks. Their goal is not only to improve accuracy on a single dataset but also to learn temporal features that remain useful across domains, tasks, and supervision levels.

This direction is driven by self-supervised pretraining, masked modeling, and large-scale time-series foundation models that enable reuse across heterogeneous datasets and tasks. Recent work further explores parameter-efficient adaptation, transferability estimation, and decomposition-based learning to reduce the cost of adapting pretrained representations to new domains. Collectively, these advances shift MTS research away from task-specific training from scratch and toward reusable temporal representations with stronger cross-domain generalization.

3.4 Axis 3: Design Capabilities (The “Why”)

Axis 3 captures the deployment-facing capabilities explicitly targeted by a method, such as scalability, robustness, interpretability, or transferability. Fig. 3 provides an illustrative summary of these capability trade-offs across model families. We assign an Axis-3 tag only when the capability is clearly reflected in the design and supported by empirical evaluation. This visualization is intended as a qualitative guide to the design space rather than as an empirical ranking of model families.

3.4.1 Scalability and Efficiency

Scalability is a central concern in MTS analysis because sequence length, sampling frequency, and sensor dimensionality can all grow rapidly. Methods such as PatchTST reduce temporal complexity through patching, while SSM-based models pursue linear-time sequence processing. Other approaches, such as MiniRocket (Dempster et al. 2021) or task-tokenized foundation models like UniTS, tackle fast inference or parameter sharing across tasks. For irregular sampling, continuous-time NCDEs (Liu et al. 2024c) avoid costly imputation by natively modeling data via differential equations. Furthermore, STG-NCDE (Choi et al. 2022) efficiently scales this continuous-time approach to spatio-temporal forecasting by incorporating physical graph topologies.

3.4.2 Robustness

Robustness in multivariate time-series modeling involves handling missing values, noise, and distribution shifts. Recent approaches improve tolerance to partial observations and spectral corruption through masked pretraining (e.g., TimeMAE) and frequency-aware contrastive learning (e.g., TimesURL). Real-world MTS systems, however, often operate in *non-stationary environments*, where data distributions evolve due to concept drift. Maintaining performance under such conditions requires models that can adapt beyond fixed representations.

Recent architectures address this challenge through *dynamic model adaptation mechanisms*. For example, memory-augmented Mixture-of-Experts (MoE) routers mitigate distribution shifts by selectively routing inputs to specialized sub-networks that capture different data regimes. These designs enable models to retain historical knowledge while responding to evolving patterns. More broadly, robust deployment in streaming environments will likely require moving beyond static architectures toward *adaptive, data-driven routing and continual learning strategies*.

3.4.3 Transferability

Recent advancements in time-series foundation models have significantly expanded the boundaries of zero-shot transferability, moving beyond traditional dataset-specific architectures. For instance, unified frameworks such as UniTS process multivariate sequences across diverse analytical tasks, including forecasting, classification, and imputation using a single shared network, demonstrating robust zero-shot and few-shot transferability across entirely unseen domains. Furthermore, advanced representation learning paradigms such as SimMTM (Dong et al. 2023) leverage masked modeling to reconstruct multivariate manifold structures, thereby extracting highly transferable features for cross-domain adaptation. However, while large-scale transformer models exhibit powerful generalization, they often incur substantial computational overhead. Addressing this efficiency gap, lightweight architectures like Tiny Time Mixers (TTM) (Ekambaram et al. 2024) leverage MLP-Mixer designs to achieve state-of-the-art zero-shot transfer in multivariate forecasting without the need for massive parameter counts. Together, these approaches highlight a dual trajectory in modern MTS research: unifying massive architectures for broad cross-task generalization, and optimizing highly efficient designs for rapid, resource-constrained adaptation.

3.4.4 Interpretability and Explainability

Interpretability remains crucial in high-stakes applications such as healthcare and finance. Existing approaches can be grouped into shape-based visual explanations, input-attribution methods, surrogate models, and natural-language explanations. Recent advancements, such as Time2Feat (Bonifati et al. 2022), extract interpretable temporal features, while Time-LLM leverages linguistic reasoning for transparency. Despite this progress, interpretability remains poorly integrated into high-performing MTS models, leaving a critical gap between predictive accuracy and real-world deployment.

3.5 Cross-Axis Relationships

The primary value of the proposed taxonomy lies in exposing the complex interactions among modeling paradigms, downstream tasks, and design capabilities. Crucially, modeling paradigms do not map one-to-one onto specific analytical tasks. For example, Transformer and state-space model (SSM) backbones are extensively utilized across forecasting, imputation, and anomaly detection. Similarly, contrastive and reconstruction-based methods increasingly support classification, transfer learning, and foundation-model pretraining. By examining these cross-axis relationships, it becomes clear that modern MTS architectures are fundamentally defined by their structural flexibility and deployment capabilities rather than being confined to a single operational domain.

Design capabilities, such as scalability, robustness, and transferability, further shape architectural choices, often as strongly as the target task itself. This many-to-many structure explains why recent MTS research is difficult to organize using task-only or architecture-only taxonomies: the same paradigm may serve multiple tasks, and the same task may require different design capabilities depending on deployment constraints.

To move beyond a theoretical discussion of these relationships, a systematic taxonomy requires explicit model-level assignments. Thus, Table 5 maps every architecture included in our reproducibility audit across all three taxonomic axes. This mapping explicitly operationalizes the taxonomy, bridging the gap between conceptual breadth and practical utility for rigorous model comparison and reproducibility analysis.

4 Reproducibility Audit of Modern MTS Models

Beyond synthesizing the literature, this survey presents a targeted reproducibility audit of representative MTS models. The audit is intended to diagnose reproducibility challenges across selected modeling paradigms and downstream tasks, rather than to provide an exhaustive benchmark of all methods or all seven tasks in the taxonomy.

Our goal is to assess whether reported performance gains in recent MTS models remain consistent under controlled experimental conditions. By replicating published results using official code repositories and standardized experimental protocols, we evaluate the robustness, transparency, and stability of modern MTS architectures.

Our experiments address the following research questions.

- **RQ1.** To what extent can reported performance in prior MTS studies be reproduced under standardized experimental settings?
- **RQ2.** How sensitive are modern MTS models to random initialization and preprocessing pipelines?
- **RQ3.** Do relative performance rankings among models remain consistent after reproduction?
- **RQ4.** How significant are the discrepancies between reproduced results and the metrics reported in the original publications?

Because the audit is targeted rather than exhaustive, the selected models are treated as representative case studies of reproducibility across major MTS paradigms. The audit spans several core tasks, including forecasting, anomaly detection, classification, clustering, and imputation, to expose recurring sources of variation across different experimental settings. Accordingly, the results are used to diagnose reproducibility barriers, not to rank the full landscape of MTS methods.

4.1 Experimental Setup

In this section, we describe the detailed configuration settings for our reproducibility evaluation.

Hardware and Software

All experiments were conducted on a system equipped with an NVIDIA Tesla V100-DGXS GPU (32 GB VRAM) running Ubuntu 20.04 LTS. The software environment included CUDA 12.2, NVIDIA Driver 535.230.02, and Python 3.12.7.

Datasets and Reproduction Protocol

To ensure a comprehensive and fair evaluation, we selected a diverse set of widely used benchmark datasets spanning the major MTS tasks considered in this survey. The datasets, summarized in Table 4, cover forecasting, anomaly detection, classification, clustering, imputation, and representation learning scenarios, enabling cross-task assessment of model behavior under varied temporal dynamics and data characteristics.

For reproducibility, we prioritized official open-source implementations whenever available. Experiments were conducted under a controlled protocol, with the random seed set to 42 to reduce stochastic variability. When implementation details such as batch size, training epochs, or learning rates were not explicitly reported in the original publications, we adopted the default configurations provided in the authors' repositories or widely accepted settings from the literature. This approach allows us to reproduce results as faithfully as possible while maintaining consistency across models.

Because certain tasks, particularly clustering, are sensitive to random initialization and optimization noise, we report clustering results averaged across three independent runs, along with the corresponding standard deviations. This protocol not only enables the verification of previously reported performance but also provides insight into the stability and reproducibility of cutting-edge deep MTS models.

Table 4 Summary of popular public MTS datasets and their typical challenges. [†]Size denotes the number of time steps.

Dataset	Size [†]	Dim	Domain	Common Challenges
PSM (Abdulaal et al. 2021)	220,322	26	Web traffic	Seasonal patterns, Sudden spikes, Rare anomalies
SMD (Su et al. 2019)	1,416,825	38	Server machine	Rare anomalies, Abrupt state shifts, High-dimensional logs
MSL (Hundman et al. 2018)	130,046	55	Satellite telemetry	Sensor drift, Irregular anomalies, Missing data
SMAP (Hundman et al. 2018)	562,800	25	Spacecraft telemetry	Multi-sensor correlations, Gradual faults, Noise
SWaT (Goh et al. 2016)	944,919	51	Industrial CPS	Cyber-physical attacks, Abrupt changes, Sensor-actuator coupling
WADI (Ahmed et al. 2017)	1,221,372	103	Water-works CPS	High dimensionality, Correlated readings, Staged attacks
ETTh (Zhou et al. 2021)	14,307	7	Energy load	Strong seasonality, Weather dependency, Long-term trends
ETTm (Zhou et al. 2021)	57,507	7	Energy load	Complex periodicity, Noise
Electricity (Trindade 2015)	26,304	321	Smart meter	Very high dimensionality, Daily cycles, Regional variation
Traffic (Lai et al. 2018)	17,451	862	Road loop sensors	Extreme dimensionality, Missing data, Rush-hour spikes
Weather (Kolle, O. et al. 2021)	52,603	21	Meteorology	Seasonal variation, Multi-var dependencies, Sensor noise
Exchange (Lai et al. 2018)	7,588	8	Finance	Non-stationarity, Volatility, Irregular shocks
Solar (National Renewable Energy Laboratory 2006)	52,560	137	Solar plant	Seasonal cycles, Cloud-cover noise, Sensor drift
ILI (Centers for Disease Control and Prevention 2025)	861	7	Epidemiology	Short series, Irregular peaks, Reporting delays
HAR (UCI) (Anguita et al. 2013)	10,299	9	Human activity	Motion variability, Sensor misalignment, Activity overlap
UCR 26 (Dau et al. 2019)	$128 \times T$	1	Modern MTS	Diverse lengths, Heterogeneous domains, Varying dims
UEA 30 (Bagnall et al. 2018)	$30 \times T$	V	Generic MTS	Extreme diversity, Class imbalance, Mixed lengths

To evaluate model performance across our taxonomy, we employ a standardized metric suite. Continuous tasks (forecasting, imputation) are assessed via Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE), alongside scale-independent MAPE and sMAPE for cross-sensor comparisons. Classification and anomaly detection utilize Accuracy and F1-score, supplemented by AUC-ROC and AUC-PR to handle severe class imbalances. Clustering quality is measured using the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) against ground truth, or the Silhouette score for label-free cohesion. Finally, representation learning and cross-domain transferability are evaluated using InfoNCE contrastive loss, reconstruction error, and Centered Kernel Alignment (CKA).

To improve transparency, we explicitly document the reproduction protocol for each dataset–model combination rather than treating the audit as a black-box rerun of

published code. For each audited combination, we record the preprocessing pipeline, normalization strategy, data-split configuration, validation policy, hyperparameter source, tuning policy, and number of random seeds. When the original paper or official repository specified these settings, we followed them directly. When details were missing, we used the default settings provided in the official implementation and explicitly reported this choice. We did not perform an extensive hyperparameter search unless it was part of the original reproduction instructions, because the goal of the audit is to *evaluate reproducibility with the information available to a typical researcher* rather than to obtain the best possible performance.

Statistical Reporting and Reproducibility Decision Rule

To make the reproducibility audit transparent and interpretable, we evaluate reproduced results not only by their deviation from the originally reported values, but also relative to the variability observed across our repeated runs. For each multi-seed experiment, we report the reproduced mean and standard deviation as $\mu_{\text{repr}} \pm \sigma_{\text{repr}}$. Let m_{orig} denote the value reported in the original publication, μ_{repr} denote the mean of our reproduced runs, and σ_{repr} denote the standard deviation across reproduced runs. Based on these quantities, each multi-seed result is assigned one of three descriptive reproducibility levels: *high*, *intermediate*, or *low*.

A result is assigned *high reproducibility* if the reproduced mean improves upon the original value, or if the original reported value falls within one standard deviation of the reproduced mean. For higher-is-better metrics, such as accuracy, precision, recall, F1, RI, and NMI, improvement means

$$\mu_{\text{repr}} \geq m_{\text{orig}}.$$

For lower-is-better metrics, such as MSE, MAE, RMSE, and anomaly score, improvement means

$$\mu_{\text{repr}} \leq m_{\text{orig}}.$$

If the reproduced mean does not improve upon the original result, the result is still classified as highly reproducible if the originally reported value lies within the one-standard-deviation range of the reproduced runs:

$$\mu_{\text{repr}} - \sigma_{\text{repr}} \leq m_{\text{orig}} \leq \mu_{\text{repr}} + \sigma_{\text{repr}}.$$

A result is assigned *intermediate reproducibility* when the reproduced mean is worse than the original value, and the original value falls outside the one-standard-deviation range but remains within two standard deviations of the reproduced mean:

$$\mu_{\text{repr}} - 2\sigma_{\text{repr}} \leq m_{\text{orig}} \leq \mu_{\text{repr}} + 2\sigma_{\text{repr}}.$$

Conversely, a result is assigned *low reproducibility* when the originally reported value falls outside the two-standard-deviation range of the reproduced runs:

$$|m_{\text{orig}} - \mu_{\text{repr}}| > 2\sigma_{\text{repr}}.$$

For example, for a higher-is-better metric such as accuracy, if the original value is 95.81% and the reproduced value is $94.34 \pm 1.05\%$, the absolute difference is 1.47. Since the two-standard-deviation threshold is $2 \times 1.05 = 2.10$, the original value falls within the two-standard-deviation range, 92.24% to 96.44%, but outside the one-standard-deviation range. We therefore assign this result *intermediate reproducibility*. In contrast, for a lower-is-better metric such as RMSE, if the original value is 11.62 and the reproduced value is 12.39 ± 0.35 , the discrepancy is 0.77. Since this exceeds the two-standard-deviation threshold of $2 \times 0.35 = 0.70$, the original value falls outside the two-standard-deviation range, 11.69 to 13.09, and the result is assigned *low reproducibility*.

Importantly, we use this 2σ rule as a *descriptive threshold* rather than as a formal statistical significance test. A rigorous paired test would require paired observations from both the original study and our reproduced runs, such as matching per-seed outputs or at least the original study’s seed-level variance. Because most original studies report only aggregate metrics without releasing per-seed results, such formal testing is generally not possible. Hence, the proposed rule provides a practical, standardized way to summarize numerical reproducibility in the context of incomplete reporting, while avoiding claims of formal statistical significance.

4.2 Reproducibility Through the Taxonomic Lens: Cross-Axis Findings

This section audits reproducibility across modern MTS architectures using the three-axis taxonomy introduced in Section 3. Rather than treating reproducibility as a single aggregate property, we examine each reproduced model according to its *modeling paradigm* (Axis 1), *downstream task* (Axis 2), and *design capability* (Axis 3).

Table 5 reports the classification for all audited models, thereby making the taxonomy operational rather than purely descriptive. To improve statistical rigor, empirical findings are reported using repeated-run averages and standard deviations whenever multi-seed reproduction is feasible, together with a predefined decision rule that distinguishes substantial reproducibility gaps from ordinary run-to-run variation.

To support this controlled baseline, Table 6 documents the experimental protocol used for each audited model, including the normalization strategy, validation split, source repository, hyperparameter origin, and random-seed setting. When a model is evaluated across multiple datasets with different preprocessing or splitting choices, dataset-specific differences are described in the corresponding audit discussion when applicable.

Building on this protocol, Section 4.2.1 examines how reproducibility varies across architectural families, including LLM-based, Transformer, graph, contrastive, diffusion, cross-modal, and state-space models, and across the downstream tasks to which they are applied. Section 4.2.2 then turns to Axis 3, analyzing how design-level factors such as computational cost, stochastic training, and implementation sensitivity affect whether reported results can be reproduced in practice. Finally, Section 4.3 summarizes recurring reproducibility barriers, including hyperparameter sensitivity, incomplete implementation details, and evaluation-protocol variability, and discusses

Table 5 Three-axis classification of the models included in the reproducibility audit.

Model	Axis 1: Paradigm	Axis 2: Task(s)	Axis 3: Capability	Reproducibility Insight
aLLM4TS	LLM-reprogramming / adaptation	Anomaly detection, classification, and imputation	Transferability; cross-task adaptation	Stable for anomaly detection at the aggregate level, but classification shows several low-reproducibility cases; imputation is mostly high or intermediate across masking ratios.
Informer	Transformer-based	Forecasting	Long-sequence scalability	Most datasets–horizon settings are high or intermediate, and selected ETTh settings exceed the predefined 2σ threshold.
FC-STGNN	Graph neural network	Forecasting, classification	Cross-variable relational modeling	Most reproduced means are worse than the reported values; selected C-MAPSS RMSE settings exceed the predefined 2σ threshold.
CSL	Contrastive representation learning	Clustering	Cluster separability	Several clustering datasets exceed the predefined 2σ reproducibility threshold, indicating sensitivity to initialization and similarity learning.
TS2Vec	Contrastive representation learning	Classification, forecasting	Transferable representations	Mostly high reproducibility for classification, but selected forecasting benchmarks exceed the predefined 2σ threshold.
ARMD	Diffusion-based generative model	Forecasting	Probabilistic temporal modeling	Several forecasting datasets exceed the predefined 2σ threshold, while selected settings match or improve upon the reported values.
TimeCMA	Cross-modal / LLM-enhanced	Forecasting	Cross-modal alignment	All audited datasets–horizon settings exceed the predefined 2σ threshold under the reproduced protocol.
TSCMamba	State-space model	Classification	Linear-time sequence modeling	Multiple classification datasets exceed the predefined 2σ reproducibility threshold, indicating dataset-specific protocol sensitivity.

their implications for high-stakes domains such as healthcare, finance, and energy

Table 6 Standardized reproduction protocol and implementation details for audited models. To report standard deviations and assess model stability, the models utilizing random seeds were evaluated over 5 independent runs. While models using a constant (official) seed value, as reported in prior work, were evaluated over 10 runs.

Model	Code Source	Hyperparam.	Normalization	Split	Runs (Seeds)
aLLM4TS	Official repo	Repo defaults	Channel-wise Z-score	Chronological 70/10/20	5 (Random)
Informer	Official repo	Paper defaults	Global standardization	Chronological 70/10/20	10 (Official)
FC-STGNN	Official repo	Repo defaults	Min-Max normalization	Random 80/20	10 (Official)
CSL	Official repo	Repo defaults	Channel-wise Z-score	N/A; unsupervised	5 (Random)
TS2Vec	Official repo	Repo defaults	Instance normalization	Chronological 70/10/20	10 (Official)
ARMD	Official repo	Paper defaults	Global standardization	Chronological 70/10/20	10 (Official)
TimeCMA	Official repo	Repo defaults	Global standardization	Chronological 70/10/20	5 (Random)
TSCMamba	Official repo	Repo defaults	Channel-wise Z-score	Random 80/20	10 (Official)

systems. Together, these perspectives show that reproducibility gaps arise not from a single source but from the interaction among architecture, task formulation, and implementation detail.

4.2.1 Architectural Families Across Task Paradigms

Our analysis shows that the empirical reliability of a model family often depends strongly on the downstream task.

LLM-Based Models (aLLM4TS)

Although recent literature reports strong performance for LLM-based adaptation across diverse time-series tasks such as, anomaly detection, classification, and imputation (Bian et al. 2024), our audit shows that reproducing these results is fundamentally task-dependent. As detailed in Table 7, the reproduced aggregate performance for anomaly detection remains close to the originally reported value. However, this aggregate agreement masks substantial dataset-metric heterogeneity under the predefined reproducibility rule. Several dataset-metric pairs are assigned *low reproducibility*, including MSL precision, recall, and F1, SMD recall, SMAP precision, and SWaT recall. For example, MSL F1 decreases from 82.26% to $79.25 \pm 0.40\%$, and SMD recall decreases from 83.09% to $80.44 \pm 1.18\%$, both exceeding the corresponding variability thresholds. In contrast, other results, such as PSM F1 and SWaT F1, remain close to or slightly improve upon the reported values. Thus, aLLM4TS appears broadly reproducible for anomaly detection at the aggregate level, but its reliability is uneven across datasets and evaluation metrics.

The classification results show a clearer reproducibility gap. As shown in Table 8, EthanolConcentration, Heartbeat, SpokenArabicDigits, and UWaveGestureLibrary are assigned *low reproducibility* because their discrepancies exceed the two-standard-deviation threshold of the reproduced runs. EthanolConcentration decreases from 33.50% to $27.00 \pm 2.36\%$, Heartbeat from 78.00% to $74.44 \pm 0.89\%$, SpokenArabicDigits

Table 7 Performance comparison for anomaly detection tasks. Bold values indicate a decrease compared to aLLM4TS (Bian et al. 2024).

Dataset	Metric	aLLM4TS	Reproduced	Reproducibility Level
SMD	P	87.87	87.42±0.65	High
	R	83.09	80.44±1.18	Low
	F1	85.42	83.78±0.82	Intermediate
MSL	P	81.58	80.68±0.29	Low
	R	82.95	77.88±0.60	Low
	F1	82.26	79.25±0.40	Low
SMAP	P	85.40	84.46±0.23	Low
	R	71.84	68.92±1.77	Intermediate
	F1	78.04	75.90±1.16	Intermediate
SWaT	P	97.90	97.92±0.01	High
	R	92.53	91.62±0.31	Low
	F1	94.57	94.67±0.16	High
PSM	P	98.47	98.63±0.08	High
	R	95.94	95.25±0.66	Intermediate
	F1	97.19	96.91±0.33	High

Table 8 Reproducibility audit for aLLM4TS (Bian et al. 2024): Classification accuracy (%). Bold indicates performance drop.

Dataset	aLLM4TS	Reproduced	Reproducibility Level
EthanolConcentration	33.50	27.00±2.36	Low
FaceDetection	68.30	64.98±2.62	Intermediate
Handwriting	27.50	27.18±0.96	High
Heartbeat	78.00	74.44±0.89	Low
JapaneseVowels	96.80	95.89±0.70	Intermediate
SpokenArabicDigits	99.00	98.59±0.12	Low
UWaveGestureLibrary	85.30	77.19±3.83	Low

from 99.00% to $98.59 \pm 0.12\%$, and UWaveGestureLibrary from 85.30% to $77.19 \pm 3.83\%$. Other datasets show weaker evidence of degradation: FaceDetection and JapaneseVowels are assigned *intermediate reproducibility* because their reproduced accuracies are lower but remain within the two-standard-deviation range, while Handwriting is assigned *high reproducibility* because its reported value, (27.50%), lies within one standard deviation of the reproduced mean ($27.18 \pm 0.96\%$). These results suggest that aLLM4TS classification is more sensitive to dataset-specific configurations, preprocessing choices, and hyperparameter settings than anomaly detection.

The imputation results are comparatively stable under the available multi-seed evaluation. As shown in Table 9, most dataset-mask-metric pairs are assigned either *high* or *intermediate reproducibility*. For example, on ETTh1 with 25% masking, the reproduced result slightly improves from 0.056 to 0.053 ± 0.006 in MSE and from 0.161 to

Table 9 Reproducibility audit for aLLM4TS (Bian et al. 2024): Imputation. Bold indicates higher error.

Dataset	Mask	aLLM4TS MSE	Reprod. MSE	Reprod. Level	aLLM4TS MAE	Reprod. MAE	Reprod. Level
ETTh1	12.5%	0.041	0.053±0.023	High	0.139	0.152±0.025	High
	25%	0.056	0.053±0.006	High	0.161	0.156±0.010	High
	37.5%	0.073	0.088±0.034	High	0.182	0.193±0.028	High
	50%	0.098	0.096±0.016	High	0.210	0.207±0.015	High
ETTh2	12.5%	0.040	0.045±0.008	High	0.126	0.136±0.013	High
	25%	0.046	0.047±0.000	Low	0.138	0.139±0.001	High
	37.5%	0.051	0.051±0.005	High	0.147	0.146±0.008	High
	50%	0.060	0.059±0.004	High	0.147	0.158±0.007	Intermediate
ETTm1	12.5%	0.019	0.021±0.006	High	0.093	0.095±0.012	High
	25%	0.025	0.022±0.001	High	0.104	0.099±0.001	High
	37.5%	0.032	0.030±0.001	High	0.118	0.113±0.002	High
	50%	0.045	0.047±0.003	High	0.139	0.142±0.005	High
ETTm2	12.5%	0.019	0.020±0.002	High	0.079	0.080±0.006	High
	25%	0.022	0.021±0.001	High	0.086	0.085±0.002	High
	37.5%	0.024	0.024±0.001	High	0.092	0.091±0.002	High
	50%	0.028	0.027±0.000	High	0.100	0.099±0.001	High

0.156 ± 0.010 in MAE. ETTh2 with 50% masking shows intermediate-level degradation, with MAE increasing from 0.147 to 0.158 ± 0.007, but the discrepancy remains within the predefined variability threshold. The only low-reproducibility case is ETTh2 with 25% masking for MSE, where the reproduced value changes from 0.046 to 0.047 ± 0.000. Because this absolute change is very small, we interpret it as a numerically minor but variance-threshold-exceeding discrepancy rather than as evidence of a practically large performance failure. Overall, aLLM4TS exhibits *task-dependent reproducibility*: anomaly detection is close at the aggregate level but heterogeneous across datasets, classification shows the clearest reproducibility gap, and imputation remains mostly stable across masking ratios.

Transformer Architectures (Informer)

Transformer-based forecasting models, represented here by Informer (Zhou et al. 2021), show generally moderate but dataset-dependent reproducibility. As shown in Table 10, most reproduced results are either close to the originally reported values, improve over them, or fall within the *intermediate reproducibility* range under the predefined decision rule. Among the 40 dataset–horizon–metric comparisons, 7 are classified as *low reproducibility*. These low-reproducibility cases are not uniformly distributed across all benchmarks; rather, they are concentrated in ETTh₁ and ETTh₂. Specifically, ETTh₁ shows low reproducibility at horizons 168 and 720 for both MSE and MAE, ETTh₂ shows low reproducibility at horizon 48 for both metrics, and ETTh₂ at horizon 168 shows low reproducibility for MAE only.

The largest discrepancies occur in the ETTh benchmarks. For ETTh₁ at horizon 168, MSE increases from 0.931 to 1.134 ± 0.091, and MAE increases from 0.752 to 0.860 ± 0.046. At horizon 720, ETTh₁ MSE increases from 1.215 to 1.408 ± 0.033, while MAE increases from 0.896 to 0.954 ± 0.015. For ETTh₂ at horizon 48, MSE increases from 1.457 to 2.408 ± 0.249, and MAE increases from 1.001 to 1.256 ± 0.079. Because these discrepancies exceed the corresponding two-standard-deviation thresholds of the reproduced runs, they are assigned *low reproducibility*.

Table 10 Reproducibility audit for Informer (Zhou et al. 2021): Comparison of reported vs. reproduced performance. Bold indicates higher error.

Dataset	Horizon	Orig. MSE	Reprod. MSE	Reprod. Level	Orig. MAE	Reprod. MAE	Reprod. Level
ETTh ₁	24	0.577	0.528±0.046	High	0.549	0.522±0.027	High
	48	0.685	0.757±0.063	Intermediate	0.625	0.673±0.033	Intermediate
	168	0.931	1.134±0.091	Low	0.752	0.860±0.046	Low
	336	1.128	1.335±0.109	Intermediate	0.873	0.943±0.045	Intermediate
	720	1.215	1.408±0.033	Low	0.896	0.954±0.015	Low
ETTh ₂	24	0.720	0.531±0.147	High	0.665	0.569±0.083	High
	48	1.457	2.408±0.249	Low	1.001	1.256±0.079	Low
	168	3.489	5.059±0.812	Intermediate	1.515	1.902±0.186	Low
	336	2.723	2.875±0.421	Intermediate	1.340	1.397±0.101	Intermediate
	720	3.467	3.965±0.421	Intermediate	1.473	1.707±0.158	Intermediate
ETTm ₁	24	0.323	0.336±0.043	Intermediate	0.369	0.384±0.032	Intermediate
	48	0.494	0.465±0.055	High	0.503	0.466±0.033	High
	96	0.678	0.559±0.059	High	0.614	0.530±0.040	High
	288	1.056	0.953±0.108	High	0.786	0.752±0.049	High
	672	1.192	1.001±0.071	High	0.926	0.789±0.053	High
Weather	24	0.335	0.335±0.010	High	0.381	0.383±0.012	Intermediate
	48	0.395	0.390±0.007	High	0.459	0.427±0.005	High
	168	0.608	0.631±0.020	Intermediate	0.567	0.587±0.014	Intermediate
	336	0.702	0.668±0.020	High	0.620	0.617±0.010	High
	720	0.831	0.659±0.022	High	0.731	0.607±0.013	High

At the same time, the concentration of low-reproducibility cases in specific datasets and horizons suggests that Informer’s reproducibility issue is not a uniform failure of the Transformer forecasting paradigm. Instead, the results indicate *sensitivity to particular benchmark configurations*, especially longer forecasting horizons and selected ETTh settings. This pattern supports a more nuanced interpretation: Informer is broadly reproducible across many dataset–horizon combinations, but its reported performance can be difficult to recover in specific settings where forecasting horizon, preprocessing, or hyperparameter choices have a stronger effect on the final error.

Graph Neural Network (FC-STGNN)

Within the family of graph neural network architectures, (Wang et al. 2024b) proposed FC-STGNN for multivariate time-series forecasting and classification. We audit its practical reproducibility using the authors’ official implementation. Table 11 compares the originally reported results with our reproduced multiple run averages and reports the corresponding reproducibility level under the predefined decision rule.

Across the evaluated FC-STGNN settings, the reproduced mean errors are consistently higher than the originally reported values, indicating worse reproduced average performance. However, not all of these numerical increases exceed the variability observed across reproduced runs. Under the predefined reproducibility decision rule, only the RMSE results on C-MAPSS FD001 and FD003 are assigned *low reproducibility*. For FD001, the reproduced RMSE increases from 11.62 ± 0.19 to 12.39 ± 0.35 , where the absolute difference, 0.77, exceeds the reproduced-run threshold $2\sigma_{\text{repr}} = 0.70$. For FD003, the reproduced RMSE increases from 11.52 ± 0.19 to 12.49 ± 0.44 , where the absolute difference, 0.97, exceeds $2\sigma_{\text{repr}} = 0.88$. Hence, these two cases meet our descriptive threshold for *low reproducibility*. In the remaining settings, the reproduced

Table 11 Reproducibility audit for FC-STGNN (Wang et al. 2024b). The table compares the originally reported results against our reproduced metrics, including classification accuracy for the UCI-HAR dataset and RMSE for the C-MAPSS datasets. Bold values indicate a drop in performance outside the 1σ threshold.

Dataset	Metric	Original	Reproduced	Reproducibility Level
C-MAPSS FD001	RMSE	11.62±0.19	12.39±0.35	Low
	Score	203±16	266±37	Intermediate
C-MAPSS FD002	RMSE	13.04±0.13	13.38±0.25	Intermediate
	Score	738±49	782±92	High
C-MAPSS FD003	RMSE	11.52±0.19	12.49±0.44	Low
	Score	198±12	280±42	Intermediate
C-MAPSS FD004	RMSE	13.62±0.25	13.91±0.27	Intermediate
	Score	816±63	913±59	Intermediate
UCI-HAR	Accuracy	95.81±0.24	94.34±1.05	Intermediate

means are worse but remain within the predefined variability range, so we interpret them as intermediate rather than definitive reproducibility failures.

The remaining rows in Table 11 are mostly assigned *intermediate reproducibility*, with the exception of the C-MAPSS FD002 Score, which achieves *high reproducibility*. In the intermediate cases, the reproduced means are worse than the originally reported values, but the discrepancies remain within the $2\sigma_{\text{repr}}$ threshold. For example, UCI-HAR accuracy decreases from $95.81 \pm 0.24\%$ to $94.34 \pm 1.05\%$, but the absolute difference of 1.47 is smaller than $2\sigma_{\text{repr}} = 2.10$. Similarly, most C-MAPSS Score metrics yield worse reproduced means while remaining within the predefined variability range. Overall, FC-STGNN shows reproducibility sensitivity in selected forecasting settings, especially FD001 and FD003 RMSE, whereas the remaining discrepancies are better interpreted as intermediate deviations rather than severe reproducibility failures.

Contrastive Representation Learning (CSL and TS2Vec)

We audit two contrastive representation learning methods: CSL for multivariate time-series clustering (Liang et al. 2023) and TS2Vec for classification and forecasting (Yue et al. 2022). Tables 12, 13, and 14 compare the originally reported results with our reproduced multi-seed and multiple run averages and assign reproducibility levels using the predefined decision rule.

For CSL, reproducibility is limited in the clustering setting. Because RI and NMI are higher-is-better metrics, a result is assigned *low reproducibility* when the originally reported value exceeds the reproduced mean by more than $2\sigma_{\text{repr}}$. Under this rule, many CSL dataset-metric pairs fall into the low-reproducibility category, with the largest discrepancies on AtrialFibrillation, StandWalkJump, NATOPS, and UWaveGestureLibrary. For example, AtrialFibrillation NMI decreases from 0.587 to 0.129 ± 0.024 , and StandWalkJump NMI decreases from 0.554 to 0.113 ± 0.007 , both far beyond the predefined threshold. However, the degradation is not universal: BasicMotions is reproduced exactly with RI and NMI of 1.000 ± 0.000 , HandMovementDirection RI is

Table 12 Comparison of CSL clustering results across datasets. Reported CSL (Liang et al. 2023) values are compared with reproduced multi-seed averages.

Dataset	Metric	Original	Reproduced	Reproducibility Level
ArticularyWordRecognition (AW)	RI	0.990	0.970±0.003	Low
	NMI	0.942	0.795±0.034	Low
AtrialFibrillation (AF)	RI	0.743	0.528±0.034	Low
	NMI	0.587	0.129±0.024	Low
BasicMotions (BM)	RI	1.000	1.000±0.000	High
	NMI	1.000	1.000±0.000	High
Epilepsy (Ep)	RI	0.873	0.784±0.025	Low
	NMI	0.705	0.554±0.040	Low
HandMovementDirection (HM)	RI	0.651	0.689±0.026	High
	NMI	0.175	0.089±0.023	Low
Libras (Li)	RI	0.941	0.934±0.001	Low
	NMI	0.761	0.724±0.004	Low
NATOPS (NA)	RI	0.876	0.802±0.002	Low
	NMI	0.657	0.401±0.023	Low
PEMS-SF (PE)	RI	0.858	0.824±0.026	Intermediate
	NMI	0.537	0.471±0.030	Low
PenDigits (PD)	RI	0.950	0.950±0.000	High
	NMI	0.822	0.811±0.011	Intermediate
StandWalkJump (SW)	RI	0.821	0.552±0.018	Low
	NMI	0.554	0.113±0.007	Low
UWaveGestureLibrary (UW)	RI	0.927	0.884±0.014	Low
	NMI	0.731	0.586±0.011	Low

classified as *high reproducibility*, and a few cases, such as PEMS-SF RI and PenDigits NMI, remain *intermediate*.

TS2Vec shows a more favorable but task-dependent pattern. In classification, most datasets are assigned *high reproducibility*, including ArticularyWordRecognition, AtrialFibrillation, BasicMotions, CharacterTrajectories, Epilepsy, JapaneseVowels, Libras, NATOPS, PEMS-SF, PenDigits, and SpokenArabicDigits, where the reproduced accuracy matches or exceeds the originally reported value. Low reproducibility is concentrated in a smaller subset of datasets, including DuckDuckGeese, EigenWorms, Heartbeat, SelfRegulationSCP1, and UWaveGestureLibrary. For instance, DuckDuckGeese decreases from 0.680 to 0.547 ± 0.035 , and SelfRegulationSCP1 decreases from 0.812 to 0.768 ± 0.011 , both exceeding the $2\sigma_{\text{repr}}$ threshold.

In forecasting, TS2Vec becomes more sensitive to the dataset and prediction horizon. All ETTh₁ horizons are classified as *high reproducibility*, and several ETTh₂ and ETTm₁ settings are also stable. In contrast, low reproducibility appears for ETTh₂ at horizons 24, 48, and 720 for both MSE and MAE, for ETTm₁ at horizon 24 for both metrics, and most prominently for Electricity, where all horizons are assigned

Table 13 Reproducibility audit for TS2Vec (Yue et al. 2022): Classification accuracy (%). Bold indicates performance drop.

Dataset	Original	Reproduced	Reproducibility Level
ArticularyWordRecognition	0.987	0.987±0.000	High
AtrialFibrillation	0.200	0.200±0.000	High
BasicMotions	0.975	0.975±0.000	High
CharacterTrajectories	0.995	0.995±0.000	High
Cricket	0.972	0.973±0.004	High
DuckDuckGeese	0.680	0.547±0.035	Low
EigenWorms	0.847	0.836±0.005	Low
Epilepsy	0.964	0.964±0.000	High
ERing	0.874	0.875±0.002	High
EthanolConcentration	0.308	0.304±0.012	High
FaceDetection	0.501	0.503±0.001	High
FingerMovements	0.480	0.488±0.022	High
HandMovementDirection	0.338	0.311±0.043	High
Handwriting	0.515	0.514±0.002	High
Heartbeat	0.683	0.669±0.004	Low
InsectWingbeat	0.466	0.465±0.001	Intermediate
JapaneseVowels	0.984	0.984±0.000	High
Libras	0.867	0.867±0.000	High
LSST	0.537	0.541±0.003	High
MotorImagery	0.510	0.508±0.019	High
NATOPS	0.928	0.930±0.003	High
PEMS-SF	0.682	0.690±0.007	High
PenDigits	0.989	0.990±0.000	High
PhonemeSpectra	0.233	0.237±0.002	High
RacketSports	0.855	0.855±0.000	High
SelfRegulationSCP1	0.812	0.768±0.011	Low
SelfRegulationSCP2	0.578	0.545±0.030	Intermediate
SpokenArabicDigits	0.988	0.988±0.000	High
StandWalkJump	0.467	0.467±0.000	High
UWaveGestureLibrary	0.906	0.903±0.001	Low

low reproducibility. For example, Electricity at horizon 24 increases from 0.287 to 1.289 ± 0.142 in MSE and from 0.374 to 0.834 ± 0.047 in MAE.

Overall, the two audits show that contrastive learning is not reproducible or irreproducible as a single category. CSL exhibits substantial reproducibility gaps in clustering; TS2Vec is mostly reproducible for classification, with several dataset-specific exceptions; and TS2Vec forecasting is sensitive to selected benchmarks, especially Electricity. These results reinforce the need to jointly evaluate reproducibility across modeling paradigms, downstream tasks, and benchmark configurations, rather than by architectural family alone.

Table 14 Reproducibility audit for TS2Vec (Yue et al. 2022): Forecasting. Bold indicates higher error.

Dataset	Horizon	Orig. MSE	Reprod. MSE	Reprod. Level	Orig. MAE	Reprod. MAE	Reprod. Level
ETTh ₁	24	0.599	0.527±0.004	High	0.534	0.500±0.002	High
	48	0.629	0.580±0.003	High	0.555	0.537±0.003	High
	168	0.755	0.738±0.003	High	0.636	0.631±0.002	High
	336	0.907	0.883±0.004	High	0.717	0.708±0.002	High
	720	1.048	1.040±0.004	High	0.790	0.789±0.001	High
ETTh ₂	24	0.398	0.415±0.004	Low	0.461	0.479±0.002	Low
	48	0.580	0.594±0.005	Low	0.573	0.592±0.002	Low
	168	1.901	1.849±0.015	High	1.065	1.063±0.004	High
	336	2.304	2.288±0.019	High	1.215	1.211±0.004	High
	720	2.650	2.737±0.014	Low	1.373	1.405±0.004	Low
ETTM ₁	24	0.443	0.496±0.010	Low	0.436	0.474±0.005	Low
	48	0.582	0.610±0.024	Intermediate	0.515	0.538±0.012	Intermediate
	96	0.622	0.621±0.013	High	0.549	0.555±0.007	Intermediate
	288	0.709	0.699±0.010	High	0.609	0.610±0.006	Intermediate
	672	0.786	0.742±0.006	High	0.655	0.636±0.003	High
Electricity	24	0.287	1.289±0.142	Low	0.374	0.834±0.047	Low
	48	0.307	1.294±0.165	Low	0.388	0.846±0.071	Low
	168	0.332	1.435±0.095	Low	0.407	0.923±0.037	Low
	336	0.349	1.395±0.097	Low	0.420	0.899±0.033	Low
	720	0.375	2.194±0.286	Low	0.438	1.173±0.081	Low

Diffusion Models (ARMD)

For the diffusion-based forecasting model ARMD (Gao et al. 2025), Table 15 compares the originally reported results with our reproduced multiple run averages. Since MSE and MAE are lower-is-better metrics, reproducibility levels are assigned according to whether the reproduced errors match, improve upon, or exceed the original values beyond the predefined variability threshold.

ARMD shows worse reproduced mean errors on most datasets, but the severity of the discrepancy varies by dataset and metric. Under the predefined decision rule, low reproducibility is observed for ETTh₁ MAE, ETTh₂ MSE and MAE, ETTm₁ MAE, ETTm₂ MSE and MAE, and Exchange MSE and MAE. The largest discrepancies appear on ETTm₂: MSE increases from 0.181 to 0.204 ± 0.001 , and MAE increases from 0.255 to 0.286 ± 0.002 , both exceeding the corresponding $2\sigma_{\text{repr}}$ thresholds. Apart from the low-reproducibility cases, ETTh₁ MSE is the only numerically worse result that remains within the predefined variability range, and is therefore interpreted as intermediate reproducibility.

By contrast, ETTh₁ MSE is assigned *intermediate reproducibility*: the reproduced error is higher than the reported value, but the discrepancy remains within the predefined variability threshold. ETTm₁ MSE is assigned *high reproducibility* because it matches the originally reported value, while both Stocks metrics achieve *high reproducibility* because the reproduced errors are lower than the reported ones. The improved Stocks results may reflect dataset-specific preprocessing or evaluation choices that are not fully specified in the original study, especially since additional Stocks-related comparisons are mentioned without complete protocol details. Overall, the ARMD audit indicates reproducibility sensitivity in selected forecasting settings, but does not support a broad claim of low reproducibility across all datasets.

Table 15 Reproducibility audit for ARMD (Gao et al. 2025): Forecasting. Bold indicates higher error.

Dataset	Metric	Original	Reproduced	Reproducibility Level
ETTh1	MSE	0.445	0.452±0.004	Intermediate
	MAE	0.459	0.464±0.002	Low
ETTh2	MSE	0.311	0.339±0.002	Low
	MAE	0.338	0.347±0.004	Low
ETTm1	MSE	0.337	0.337±0.000	High
	MAE	0.376	0.385±0.001	Low
ETTm2	MSE	0.181	0.204±0.001	Low
	MAE	0.255	0.286±0.002	Low
Exchange	MSE	0.093	0.099±0.002	Low
	MAE	0.203	0.222±0.004	Low
Stocks	MSE	0.235	0.075±0.003	High
	MAE	0.269	0.189±0.004	High

Table 16 Reproducibility audit for TimeCMA (Liu et al. 2024a): Forecasting. Bold indicates a worse reproduced mean; the final column reports whether the discrepancy exceeds the predefined variability threshold.

Dataset	Horizon	Orig. MSE	Reprod. MSE	Reprod. Level	Orig. MAE	Reprod. MAE	Reprod. Level
ETTh1	96	0.373	0.391±0.001	Low	0.391	0.413±0.000	Low
	192	0.427	0.432±0.000	Low	0.421	0.435±0.000	Low
	336	0.458	0.469±0.002	Low	0.448	0.452±0.001	Low
	720	0.449	0.463±0.002	Low	0.460	0.466±0.001	Low
ETTh2	96	0.286	0.330±0.005	Low	0.336	0.365±0.003	Low
	192	0.372	0.407±0.007	Low	0.389	0.412±0.005	Low
	336	0.401	0.440±0.003	Low	0.406	0.440±0.001	Low
	720	0.427	0.445±0.006	Low	0.416	0.453±0.004	Low
ETTm1	96	0.312	0.324±0.001	Low	0.351	0.365±0.001	Low
	192	0.361	0.374±0.001	Low	0.378	0.393±0.000	Low
	336	0.392	0.406±0.001	Low	0.401	0.414±0.000	Low
	720	0.453	0.469±0.001	Low	0.438	0.448±0.000	Low
ETTm2	96	0.173	0.182±0.002	Low	0.258	0.264±0.003	Low
	192	0.238	0.254±0.002	Low	0.301	0.312±0.004	Low
	336	0.297	0.313±0.002	Low	0.338	0.350±0.002	Low
	720	0.333	0.415±0.005	Low	0.394	0.407±0.004	Low

Cross-Modal Architectures (TimeCMA)

For the cross-modal forecasting model TimeCMA (Liu et al. 2024a), Table 16 compares the originally reported errors with our reproduced multi-seed averages. Across all evaluated datasets, horizons, and metrics, the reproduced MSE and MAE values are higher than the reported values. Since both metrics are lower-is-better, this indicates worse reproduced mean performance; the predefined reproducibility level then determines whether each discrepancy exceeds ordinary run-to-run variability.

The largest discrepancies appear on ETTh₂. Reproduced MSE increases from 0.286 to 0.330 ± 0.005 at horizon 96, from 0.372 to 0.407 ± 0.007 at horizon 192, and from

0.401 to 0.440 ± 0.003 at horizon 336. MAE shows a similar pattern, with ETTh₂ at horizon 720 increasing from 0.416 to 0.453 ± 0.004 . ETTm₂ also shows consistent discrepancies, including an MSE increase from 0.393 to 0.415 ± 0.005 at horizon 720. These differences exceed the corresponding predefined variability thresholds and are therefore assigned low reproducibility.

Overall, the TimeCMA audit indicates substantial reproducibility challenges for this specific cross-modal architecture under our reproduced protocol. The consistent direction of the error increases suggests sensitivity to implementation or preprocessing details, particularly normalization and dataset preparation choices that may not be fully specified. Hence, we interpret these findings as evidence of low reproducibility for the audited TimeCMA settings, rather than as a general conclusion about all cross-modal MTS methods.

State-Space Models (TSCMamba)

For the state-space classification model TSCMamba, Table 17 compares the originally reported accuracies with our reproduced multiple run averages. The reproduced accuracies are lower than the reported values for most datasets, and many discrepancies exceed the predefined variability threshold, resulting in *low reproducibility* assignments.

The largest gaps occur on EigenWorms, EthanolConcentration, Handwriting, LSST, PhonemeSpectra, and DuckDuckGeese. For example, EigenWorms decreases from 87.00% to $42.824 \pm 1.414\%$, and Handwriting decreases from 53.30% to $21.976 \pm 1.318\%$, both far beyond the $2\sigma_{\text{repr}}$ threshold. In contrast, JapaneseVowels and ERing are assigned *high reproducibility*: JapaneseVowels slightly improves upon the reported value, while ERing remains within the one-standard-deviation range. A few datasets, including ArticulatoryWordRecognition, AtrialFibrillation, and PenDigits, fall into *intermediate reproducibility*. The InsectWingbeat result is omitted because the dataset was computationally expensive to process under the available preprocessing pipeline.

Overall, the TSCMamba audit indicates substantial reproducibility challenges for several classification benchmarks, but we avoid attributing these gaps solely to random-seed sensitivity. The discrepancies may also reflect dataset-specific preprocessing, hyperparameter settings, or implementation differences between the original and reproduced protocols. The omitted *InsectWingbeat* case further shows that reproducibility can be constrained not only by final model performance, but also by preprocessing cost and computational feasibility.

4.2.2 Design Attributes: Computational Efficiency and Algorithmic Stability

The third axis of our taxonomy design capabilities reveals additional barriers to practical deployment. First, computational cost remains a major limitation; for example, the substantial requirements of aLLM4TS prevented full reproduction of its forecasting experiments under standard hardware. Moreover, hardware generalization issues often arise when architectural gains are tied to specific kernel optimizations rather than portable designs.

Second, stochastic training dynamics can significantly influence reported results. Architectures like Informer and CSL exhibited pronounced sensitivity to random

Table 17 Reproducibility audit for TSCMamba (Ahamed and Cheng 2025): Classification accuracy (%). Bold indicates performance drop.

Dataset	Original	Reproduced	Reproducibility Level
ArticularyWordRecognition	97.00	96.567±0.903	Intermediate
AtrialFibrillation	67.00	52.667±7.981	Intermediate
BasicMotions	100.00	86.500±1.291	Low
CharacterTrajectories	99.00	98.628±0.143	Low
Cricket	98.61	93.472±2.076	Low
DuckDuckGeese	62.00	38.200±3.327	Low
EigenWorms	87.00	42.824±1.414	Low
Epilepsy	97.10	72.246±3.056	Low
ERing	91.50	90.037±2.650	High
EthanolConcentration	62.00	28.897±0.948	Low
FaceDetection	69.40	63.130±1.484	Low
FingerMovements	69.00	55.700±3.860	Low
HandMovementDirection	71.62	53.378±3.888	Low
Handwriting	53.30	21.976±1.318	Low
Heartbeat	76.60	74.488±0.566	Low
InsectWingbeat	14.30	N/A	Not available
JapaneseVowels	97.00	97.757±0.585	High
Libras	90.00	79.778±1.554	Low
LSST	60.30	31.967±0.516	Low
MotorImagery	62.00	57.000±1.886	Low
NATOPS	94.44	88.222±1.194	Low
PEMS-SF	90.20	78.035±3.844	Low
PenDigits	98.54	98.293±0.130	Intermediate
PhonemeSpectra	24.66	9.448±0.378	Low
RacketSports	91.45	80.658±1.461	Low
SelfRegulationSCP1	92.50	89.317±0.911	Low
SelfRegulationSCP2	66.70	56.722±1.094	Low
SpokenArabicDigits	99.00	98.022±0.280	Low
StandWalkJump	73.33	56.000±7.166	Low
UWaveGestureLibrary	93.80	86.656±0.955	Low

initialization, while TSCMamba showed a strong dependence on dataset-specific configurations, highlighting challenges in SSM scalability. These observations suggest that reported performance gains may be influenced by experimental variance rather than purely architectural improvements.

In summary, our findings indicate that although modern MTS architectures achieve strong performance, their empirical reliability depends critically on experimental design choices. Improving reproducibility requires greater transparency in preprocessing pipelines, multi-seed evaluation protocols, and clearer reporting of computational settings. Addressing hardware generalization and scalability is therefore essential to ensure reliable and reproducible progress in MTS research.

Table 18 Summary of key reproducibility challenges identified in our empirical audit of modern MTS models

Challenge	Observed Issues in Reproduction	Implications for MTS Research
<i>Hyperparameter Sensitivity</i>	Performance varied significantly with changes in batch size, training epochs, or optimizer settings. In several cases, the exact configuration used in the original work was not fully described.	Reproducing reported results becomes difficult; model performance may depend heavily on careful tuning rather than architectural improvements.
<i>Incomplete Implementation Details</i>	Critical experimental steps, such as data normalization, padding strategies, dataset splits, and preprocessing pipelines, were sometimes undocumented.	Small differences in preprocessing can lead to measurable performance gaps, rendering cross-paper comparisons unreliable.
<i>Residual Stochasticity</i>	Even with identical hyperparameters and fixed random seeds, results occasionally differed due to nondeterministic GPU operations, data-loading order, or hardware differences.	Deep learning experiments may exhibit unavoidable variance, which indicates that reporting single-run results is insufficient.
<i>Random Seed Sensitivity</i>	Certain models showed large performance fluctuations across random seeds. In clustering experiments, scores varied substantially across initializations.	Published results may result from favorable seeds rather than stable performance. Multi-run reporting is therefore essential.
<i>Computational Constraints</i>	Some long-horizon forecasting experiments required multiple high-memory GPUs and several days of training.	High computational cost limits reproducibility and makes systematic hyperparameter exploration impractical for many researchers.
<i>Evaluation Protocol Variability</i>	Studies frequently used different dataset splits, prediction horizons, scaling procedures, or metric definitions.	Differences in evaluation protocols can lead to large numerical discrepancies, complicating fair comparisons across models.

4.3 Lessons

Table 18 summarizes the primary factors influencing reproducibility in our audit. Discrepancies between reported and reproduced results often stem not from fundamental design flaws, but from under-specified experimental protocols, stochastic training dynamics, and heterogeneous evaluation settings.

These barriers are especially critical in real-world deployments. MTS models increasingly drive decision-making in high-stakes domains, where training instability or ambiguous evaluation protocols translate directly into operational risks. Our audit further indicates that these impacts are highly domain-dependent:

Safety-Critical Instability (Healthcare and Manufacturing)

These applications demand highly reliable predictions from noisy sensor data. The sensitivity to initialization and training conditions observed in our audit raises concerns for safety-critical settings, where inaccurate predictions could delay clinical interventions or obscure equipment failures.

Preprocessing Ambiguity (Finance and Retail)

Strong reported performance in financial tasks often depends on highly specific, yet incompletely documented, preprocessing pipelines (e.g., normalization and feature filtering). In operational environments, these hidden dependencies complicate reproduction and increase the risk of unpredictable performance degradation.

Evaluation Protocol Gaps (Energy Systems)

While energy systems must handle continuously evolving temporal dynamics, many studies rely on simplified fixed train-test splits rather than rolling-window or streaming evaluations. Such choices can overestimate model stability and obscure real-world performance variability.

Overall, reproducibility challenges in MTS research arise from stochastic training dynamics, incomplete reporting, and inconsistent evaluation protocols cutting across all architectural paradigms in our taxonomy. Addressing these open concerns requires greater experimental transparency, standardized benchmarking, and routine multi-run reporting to accurately characterize model stability across frameworks.

5 Research Challenges and Complexities

Multivariate time-series (MTS) analytics face several intrinsic challenges that affect not only model accuracy but also the *comparability*, *deployability*, and *reproducibility* of proposed methods. These challenges arise from the complex structure of MTS data, the computational demands of modern deep architectures, and inconsistencies in experimental evaluation protocols.

In this section, we summarize the most consequential challenges observed across the literature and in our reproducibility audit (Section 4). Rather than focusing on open research questions to be discussed in the next section, we highlight practical failure modes that frequently arise when applying deep learning models to real-world MTS data.

5.1 Challenges in MTS Data

Modern applications generate massive volumes of multivariate time-series data from sensors, industrial systems, financial markets, and healthcare monitoring. The rapid growth of such data, estimated to reach tens of zettabytes annually from IoT devices alone (Reinsel et al. 2019), has shifted MTS analysis from classical statistical models to data-driven deep learning approaches. However, the scale and complexity of these datasets introduce several structural difficulties.

5.1.1 Spatio-Temporal Dependency Modeling

A central challenge in MTS analysis is capturing both the temporal evolution within each variable and the interactions among variables.

Temporal Dynamics

Individual variables often exhibit trends, seasonal cycles, and irregular fluctuations. Recurrent neural networks, such as LSTMs, were designed to model these dynamics but remain *vulnerable* to vanishing gradients and limited long-range memory. As forecasting horizons grow, small modeling errors can accumulate, significantly degrading predictive accuracy.

Cross-Variable Dependencies

In many systems, variables influence one another through physical or logical relationships. As the number of variables increases, the number of potential interactions grows quadratically, creating a high-dimensional modeling problem (Zhang and Yan 2023). Furthermore, dependencies may arise with time delays rather than synchronous observations, leading models to misinterpret natural lagged relationships as anomalies.

5.1.2 Data Quality: Missingness and Noise

Real-world MTS data are seldom clean. Sensors may fail, sampling may be irregular, and measurements may contain substantial noise.

Imputation–Prediction Coupling

A common strategy is to impute missing values before applying predictive models. However, because the true values are unknown, imputation errors can propagate into downstream predictions. Moreover, many imputation techniques assume data are missing at random, which is often unrealistic in domains such as healthcare.

Informative Missingness

In many settings, the absence of an observation itself carries meaningful information. For example, a clinician may omit a test because the patient appears stable. Treating such missing values as simple noise may erase important predictive signals. As a result, models must often reason directly with partially observed sequences rather than relying solely on preprocessing-based imputation.

5.1.3 Distribution Shift and Non-Stationarity

The statistical properties of real-world time series frequently evolve over time, leading to distribution shifts that degrade model performance.

- **Covariate shift:** the distribution of input variables changes over time.
- **Concept drift:** the relationship between inputs and target outputs evolves due to environmental changes, system aging, or user behavior (Gama et al. 2014).
- **Spectral variability:** periodic patterns may change frequency or amplitude, making static frequency-based representations unreliable.

Standard mitigation techniques, such as normalization or detrending, can stabilize training but may also remove long-term dependencies that are crucial for accurate prediction. Furthermore, conventional static train–test splits often underestimate the severity of distribution shifts in real deployment settings.

5.1.4 Multi-Scale Dynamics and Computational Cost

MTS data often contain patterns operating at multiple temporal scales, such as short-term fluctuations embedded within longer seasonal cycles. Capturing these hierarchical dynamics is computationally demanding.

Transformer-based models can model long-range dependencies but exhibit quadratic computational complexity with respect to sequence length. Although recent architectures attempt to reduce this cost, long-horizon modeling of high-dimensional MTS data remains computationally challenging, particularly in real-time or resource-constrained environments such as edge devices (Jang et al. 2024, 2025).

5.1.5 Channel Modeling Strategies

A fundamental design choice in deep MTS models concerns how variables (or *channels*) should be processed.

Channel-Independent Modeling

Some architectures treat each variable independently, thereby reducing model complexity and improving generalization in high-dimensional settings. That said, this approach disregards potentially important interactions among variables.

Channel-Dependent Modeling

Other models explicitly learn the relationships between variables, often through attention mechanisms or graph structures. While this strategy can reflect richer dependencies, it increases model complexity and may further amplify noise when interactions are weak or poorly estimated.

Balancing these two strategies poses a persistent architectural challenge.

5.1.6 Interpretability and Causal Understanding

As MTS models are increasingly used in high-stakes domains such as healthcare and finance, interpretability is more imperative than ever.

Deep learning models excel at identifying statistical correlations but do not inherently distinguish causal relationships from spurious patterns (Runge et al. 2019). This limitation can hurt trust in model predictions and complicate decision-making in safety-critical environments. Furthermore, many high-performing architectures sacrifice transparency for predictive power, creating a tension between model complexity and interpretability.

5.1.7 Benchmarking and Evaluation Integrity

Ensuring fair evaluation has been nontrivial as datasets and pretrained models grow larger. In particular, large time-series foundation models may be trained on extensive public datasets, making it hard to guarantee that benchmark test sets were not indirectly included during pretraining. Such leakage can artificially inflate reported performance and complicate comparisons across studies.

5.1.8 Multimodal Integration

Recent studies tend to combine time-series data with other modalities such as text, images, or structured metadata. Although multimodal integration can improve predictive performance, aligning heterogeneous data sources remains difficult because modalities differ in temporal resolution, semantics, and information density. Naive fusion strategies may obscure significant temporal patterns or incur spurious correlations.

5.2 Complexities in Downstream Tasks

Beyond data-related challenges, additional complexities arise when evaluating models on downstream MTS tasks. In many cases, disparities in experimental protocols, such as dataset splits, preprocessing pipelines, and metric definitions, may yield large performance discrepancies across studies.

5.2.1 Anomaly Detection

Anomaly detection is especially sensitive to evaluation design, as anomalies are rare and often span multiple variables. Reported performance can vary drastically depending on thresholding strategies, tolerance windows, or point-adjustment techniques (Sørbo and Ruocco 2023; Paparrizos et al. 2022; Liu and Paparrizos 2024). These protocol choices may produce bubble scores and lead to misleading comparisons between models.

5.2.2 Classification

MTS classification often suffers from limited labeled data, class imbalance, and heterogeneous sequence lengths. Deep learning models can capture nonlinear dependencies but may require careful regularization and augmentation to avoid overfitting. Transparent reporting of data splits, hyperparameters, and performance variance is thus indispensable for reproducibility.

5.2.3 Clustering

MTS clustering remains challenging because unsupervised objectives may not align with practical utility. Results can also be highly sensitive to initialization, similarity definitions, and random seeds. Our audit demonstrates that clustering performance should be reported *across* multiple runs with a clear specification of distance metrics and embedding strategies.

5.2.4 Forecasting

Forecasting performance is strongly influenced by experimental design choices such as context window length, normalization methods, and prediction horizons. Small variations in preprocessing or evaluation protocols can produce systematic performance differences across studies. Additionally, many forecasting studies focus solely on point predictions, failing to report predictive uncertainty, which is of paramount importance in decision-making settings.

5.2.5 Imputation

Imputation tasks must preserve temporal dependencies and cross-variable relationships under varying missingness mechanisms. As many papers evaluate imputation under synthetic masking schemes, the robustness of these models to real-world missingness patterns is still unresolved.

5.2.6 Representation Learning and Transfer

Self-supervised and transfer learning approaches promise improved generalization across tasks, but they pose new challenges in evaluation. Representation quality is often measured indirectly through downstream tasks, yielding results that depend on fine-tuning protocols, dataset choices, and transfer settings. Negative transfer can also occur when pretraining distributions differ substantially from those of the target domains.

All in all, these challenges illustrate that progress in MTS research continues not only with improved model architectures but also with careful attention to data properties, evaluation protocols, and deployment constraints.

6 Open Research Gaps and Future Directions

This section highlights underexplored challenges and promising future directions in MTS analysis. While deep learning has advanced the field, significant gaps persist in representation learning, multimodal integration, clustering, imputation, streaming adaptation, and unified anomaly modeling.

Importantly, the future directions discussed below are directly motivated by the empirical failure modes observed in our reproducibility audit (Section 4.2). The audit reveals several distinct vulnerabilities in modern MTS architectures: (1) LLM-based and cross-modal models exhibit sensitivity to preprocessing, modality alignment, and dataset preparation; (2) state-space models such as TSCMamba show strong task-specific sensitivity in classification; (3) contrastive clustering outcomes vary substantially across random seeds and benchmark settings; and (4) imputation performance in models such as aLLM4TS can vary across masking configurations. Thus, we explicitly connect each research gap and future direction below to the corresponding empirical failure observed in the audit.

Table 19 summarizes the main research gaps, the limitations of current approaches, and future pathways. These challenges are not isolated; they reflect a broader methodological need for robustness, multimodal reasoning, streaming adaptation, principled similarity learning, and realistic evaluation. We now discuss each of these gaps in detail.

6.1 Open Research Gaps

While deep learning has significantly advanced MTS analysis, critical gaps remain in representation learning, multimodal integration, clustering, imputation, and streaming adaptation. These intertwined challenges reflect a broader need for robustness, dynamic adaptation, and realistic evaluation under noisy, real-world conditions.

Despite recent progress, we identify several open research questions that remain insufficiently addressed:

Table 19 Open research gaps in MTS analysis, their current methodological limitations, and future directions linked to the research questions and reproducibility failures identified in Section 4.2.

Research Gap	Why Current Methods Fall Short	Promising Direction
<i>Generalizable and robust representation learning</i>	Self-supervised methods rely on static augmentations that distort patterns under domain shift. The audit reveals that representation quality is highly sensitive to the downstream task and evaluation protocol.	<i>Robust and Task-Adaptive Representation Learning:</i> Develop stability-aware objectives that preserve temporal semantics across tasks/seeds, incorporating uncertainty and task-adaptive representation selection.
<i>Multimodal fusion for MTS</i>	Extant pipelines encode modalities separately and rely heavily on undocumented preprocessing. As shown in the audit, small changes in tokenization or scaling lead to non-trivial reproduction gaps.	<i>Transparent LLM and Multimodal Adaptation:</i> Build transparent, theoretically grounded adaptation pipelines that explicitly document tokenization, prompt design, and alignment objectives to ensure genuine cross-modal fusion.
<i>Similarity learning for MTS clustering</i>	Pipelines lack metrics that jointly capture temporal alignment and cross-channel covariance. Forecasting-oriented models fail to provide similarity measures suitable for unsupervised grouping.	<i>Stable and Clustering-Oriented Deep MTS Models:</i> Formulate alignment- and covariance-aware similarity metrics specifically tailored for clustering, moving beyond metrics designed solely for forecasting.
<i>Clustering-oriented deep MTS architectures</i>	Models often treat clustering as an indirect byproduct of forecasting. As the audit demonstrated (e.g., CSL), this leads to extreme instability and varying outcomes across different random seeds.	<i>Stable and Clustering-Oriented Deep MTS Models:</i> Construct architectures with explicit clustering-first objectives that penalize seed-sensitive solutions, reporting variance rather than single-run point estimates.
<i>Imputation under NMAR missingness</i>	Models often assume random missingness and rely on synthetic uniform masking. The audit shows that imputation reproducibility can vary across masking configurations, motivating evaluation protocols that account for value-dependent missingness.	<i>Missingness-Aware and NMAR-Aware Imputation:</i> Explicitly model the missingness mechanism (NMAR) instead of assuming random drops, pairing these objectives with calibrated uncertainty estimation.
<i>Streaming adaptation and concept drift</i>	Conventional models detect drift late or rely on static training protocols that fail in non-stationary environments. High computational costs hinder real-time deployment under strict latency limits.	<i>Streaming and Real-Time Adaptation:</i> Implement lightweight online adaptation frameworks with reproducibility-aware evaluation (e.g., rolling-window validation) to verify stability under dynamic data shifts.
<i>Unified spatiotemporal anomaly modeling</i>	Methods typically treat temporal and cross-channel anomalies independently using static structures, struggling with irregular sampling and failing to generalize across different anomaly types.	<i>Robust and Task-Adaptive Representation Learning:</i> Build unified frameworks that jointly reason over temporal and cross-channel irregularities dynamically, integrating uncertainty to improve interpretability and reliability.

- **OP1: How can MTS representations be made both generalizable and robust?** While TS2Vec (Yue et al. 2022) performs well, static augmentations can distort patterns under domain shift (Um et al. 2017). Given Transformer computational overhead, adaptive and efficient representation strategies are required for reliable cross-domain transfer.
- **OP2: How can temporal representations be fused effectively with heterogeneous modalities?** Current pipelines typically encode complementary modalities (e.g., text, tabular) separately, failing to align heterogeneous spaces. Future frameworks must jointly model temporal structure, inter-modality dependencies, and uncertainty.
- **OP3: How should similarity be defined for MTS clustering?** Defining similarity that respects temporal alignment and cross-channel dependencies remains

difficult. Methods like FCPCA (Ma et al. 2025) miss nonlinearities, while forecasting models (DUET (Qiu et al. 2024), DGCformer (Liu et al. 2024b)) lack clustering-oriented metrics, necessitating novel covariance-aware measures.

- **OP4: How can clustering become a primary objective in deep MTS models?** Models like ForecastGrapher (Cai et al. 2024a) and Temporal Graph Clustering (Liu et al. 2023a) treat clustering as an indirect byproduct of forecasting. Future work must establish explicit objectives to directly optimize representations for unsupervised grouping.
- **OP5: How can imputation handle value-dependent missingness?** Models (MissNet (Obata et al. 2024), FGTI (Yang et al. 2024), SAITS (Du et al. 2023)) typically assume random missingness. To avoid bias in *not missing at random* (NMAR) scenarios, methods must explicitly model the missingness process alongside calibrated uncertainty.
- **OP6: How can streaming MTS models adapt to concept drift under strict latency constraints?** Approaches like OneNet (Wen et al. 2023) and Proceed (Zhao and Shen 2025) address drift but often detect it late or require expensive continuous-time models. Developing lightweight architectures that adapt to online drift without catastrophic forgetting remains a critical challenge.
- **OP7: How can temporal and cross-channel anomalies be modeled jointly?** Most methods tackle temporal irregularity and cross-channel inconsistency independently. Although models like STGCN-Hyper (Xu 2024) attempt joint modeling, their reliance on static structures highlights the need for unified, adaptive spatiotemporal reasoning.

6.2 Future Directions

The research gaps identified above should be interpreted in light of the empirical failure modes observed in the reproducibility audit. Rather than presenting future work as a generic list of open problems, we connect each direction to concrete patterns observed in Section 4.2, including sensitivity to preprocessing, random initialization, masking schemes, evaluation protocols, and task-specific experimental design.

Robust and Task-Adaptive Representation Learning.

Addressing the need for generalizable representations (OP1) and unified spatiotemporal anomaly modeling (OP7) requires methods that go beyond static augmentations and task-isolated modeling. The audit indicates that representation quality may vary substantially across downstream tasks and evaluation protocols. For example, contrastive representations were relatively stable under supervised classification, whereas clustering performance was more sensitive to initialization and similarity structure. Likewise, *state-space models* exhibited large reported-reproduced gaps on several classification datasets. These findings imply that future representation-learning methods should optimize not only average downstream accuracy, but also stability across random seeds, task formulations, and dataset characteristics. Promising directions include stability-aware contrastive objectives, task-adaptive representation selection, and unified anomaly-detection frameworks that jointly model temporal dynamics and cross-channel dependencies under uncertainty.

Missingness-Aware and NMAR-Aware Imputation.

Overcoming the limitations of value-independent missingness assumptions (**OP5**) requires imputation models that explicitly account for the mechanisms underlying missingness. The audit shows that imputation performance can depend on masking configuration. The reproduced *aLLM4TS* results indicate that imputation performance is generally stable under the revised multi-seed audit, but can still vary across masking configurations. This suggests that current evaluation protocols remain sensitive to how missing values are generated and assessed. In real-world settings such as healthcare, industrial sensing, and environmental monitoring, missingness is often not missing at random (NMAR); the absence of a measurement may itself encode informative system states. Future work should therefore develop NMAR-aware objectives, calibrated uncertainty estimates, and benchmarks with realistic missingness patterns rather than relying primarily on uniform or artificially generated masking schemes.

Stable and Clustering-Oriented Deep MTS Models.

Developing robust covariance-aware similarities (**OP3**) and establishing clustering as a primary objective (**OP4**) require moving beyond the common practice of treating clustering as a secondary evaluation of representations learned for other tasks. The *CSL* reproduction results show that clustering can be highly unstable across random seeds, even when using the official implementation. Future deep MTS clustering methods should therefore directly optimize cluster separability, temporal coherence, and cross-channel structure while reducing sensitivity to initialization. In addition, multi-run reporting should become standard for clustering experiments. Reporting only a single run can substantially overstate reliability, whereas mean and standard deviation across repeated runs provide a more faithful picture of empirical stability.

Transparent LLM and Multimodal Adaptation Protocols

Effectively integrating temporal sequences across heterogeneous modalities (**OP2**) requires transparent, theoretically grounded adaptation pipelines. The audit shows that LLM-based and cross-modal time-series methods can be highly sensitive to preprocessing, tokenization, normalization, patch construction, and alignment choices. For instance, *TimeCMA* exhibited substantial preprocessing sensitivity associated with undocumented normalization protocols. Because these methods transform continuous temporal values into token-like, patch-based, or cross-modal representations, seemingly minor implementation choices can produce nontrivial reproduction gaps. Future work on LLM reprogramming and multimodal MTS modeling should therefore report the full adaptation pipeline, including scaling procedures, patch construction, prompt design, validation selection, modality alignment objectives, and inference settings. Such transparency is necessary to determine whether performance gains arise from the proposed multimodal architecture itself or from undocumented preprocessing and tuning choices.

Streaming and Real-Time Adaptation Under Reproducibility Constraints.

Enabling streaming models to adapt to concept drift under latency and resource constraints (**OP6**) requires lightweight online learning together with reproducibility-aware evaluation. The task-specific reproduction gaps observed for several models suggest that static train–test protocols may not adequately reflect deployment in non-stationary environments. Future streaming MTS models should therefore combine online adaptation with rigorous evaluation protocols, including rolling-window validation, explicit concept-drift detection, low-latency update mechanisms, and transparent reporting of computational budgets. These practices are essential for assessing whether a model remains stable as the data distribution changes over time, rather than only under a fixed offline split.

Rigorous Reproducibility and Evaluation Standards.

Progress across these directions ultimately depends on stronger evaluation standards. The audit shows that reproduction gaps often arise from incomplete reporting of preprocessing steps, validation splits, hyperparameters, random seeds, implementation details, and hardware settings. Future MTS benchmarks should therefore require full preprocessing scripts, dataset versioning, multi-seed evaluation, and disclosure of the compute budget. For repeated experiments, results should be reported with mean and standard deviation, and substantial discrepancies should be assessed using transparent statistical criteria, such as whether the reported–reproduced difference exceeds two standard deviations of reproduced runs. These standards are necessary to distinguish genuine methodological progress from favorable random seeds, undocumented tuning, or dataset-specific preprocessing choices.

All in all, these directions suggest that the next generation of MTS research must move beyond isolated task-specific performance gains toward models that are robust, adaptive, interpretable, and reproducible under realistic deployment conditions.

7 Limitations

While this survey aims to provide a comprehensive synthesis of recent advances in multivariate time-series modeling, several limitations exist.

First, the survey deliberately focuses on deep representation-learning approaches. Classical similarity-based, statistical, and indexing methods, such as elastic distance measures, shape-based techniques, and matrix-profile-based frameworks, have played a crucial role in the broader time-series literature. These approaches are mainly discussed for historical context rather than analyzed exhaustively. Accordingly, the survey should be interpreted as a deep-learning-centric perspective on modern multivariate time-series research rather than as a complete survey of all methodological paradigms.

Second, the reproducibility audit is intentionally limited in scope. While the taxonomy provides broad organizational coverage of seven MTS task families, the audit is designed as a targeted empirical study rather than a comprehensive benchmark. It evaluates a representative subset of models selected to illustrate reproducibility behavior across several major modeling paradigms and downstream tasks, but it does not exhaustively cover all MTS architectures, datasets, or taxonomy categories. In particular, not

every task family in the taxonomy is evaluated as a standalone experimental setting. Accordingly, the audit results should be interpreted as evidence of recurring reproducibility challenges, such as sensitivity to preprocessing, hyperparameter specification, random seeds, implementation details, and dataset protocols, rather than as a complete ranking or definitive benchmark of modern MTS methods. Expanding this empirical evaluation to a broader range of architectures and standardized experimental protocols remains an important direction for future community-driven benchmarking efforts.

Lastly, the rapid pace of progress in time-series modeling, particularly with the emergence of foundation models and large-scale pretraining strategies, means that new architectures and training paradigms are likely to appear after the completion of this survey. Although our taxonomy intends to be extensible, future developments may introduce new modeling principles that require further refinement or expansion of the framework.

8 Conclusion

In this manuscript, we presented a *deep learning-centric* survey of multivariate time series (MTS) analytics, grounded with essential classical building blocks and organized around seven downstream tasks, delivering three primary contributions: First, we introduced a *multi-dimensional taxonomy* that categorizes modern MTS methods by their modeling paradigms, task objectives, and practical capabilities. This framework reveals how common architectures are shared across different tasks and clarifies the inevitable trade-offs between accuracy, efficiency, and robustness.

Second, we conducted a *targeted reproducibility audit* of representative open-source models. The audit was not intended to provide an exhaustive benchmark across all models or all taxonomy tasks; rather, it uses selected representative models to illustrate how incomplete preprocessing details, hyperparameter sensitivity, stochastic training, and inconsistent evaluation protocols can affect reproduced results.

Finally, we synthesized *persistent challenges* in real-world MTS deployments, including non-stationarity, missing data, and concept drift. To overcome these, future models must prioritize interpretability, uncertainty awareness, and the integration of domain-specific knowledge. Ultimately, we strongly advocate for stricter community standards, including dataset versioning, transparent preprocessing scripts, multi-run variance reporting, and clear compute budgets, to ensure reliable, reproducible progress in MTS research.

Acknowledgment

This work was supported by the National Research Foundation of Korea (NRF) grants (RS-2025-16067408 and RS-2018-NR031059) and by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant (RS-2025-25453740, Development of a large-scale mixed device control and management platform for edge AI server systems) funded by the Korean government (MSIT).

References

- Abdulaal A, Liu Z, Lancewicki T (2021) Practical Approach to Asynchronous Multivariate Time Series Anomaly Detection and Localization. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining, pp 2485–2494
- Aghabozorgi S, et al (2015) Time-series clustering—a decade review. *Information Systems* 53:16–38
- Ahamed MA, Cheng Q (2024) Timemachine: A time series is worth 4 mambas for long-term forecasting. In: ECAI 2024: 27th European Conference on Artificial Intelligence, 19-24 October 2024, Santiago de Compostela, Spain-Including 13th Conference on Prestigious Applications of Intelligent Systems. European Conference on Artificial Intelli, p 1688
- Ahamed MA, Cheng Q (2025) Tscmamba: Mamba meets multi-view learning for time series classification. *Information Fusion* 120:103079. <https://doi.org/10.1016/j.inffus.2025.103079>
- Ahmed CM, Palleti VR, Mathur AP (2017) WADI: A water Distribution Testbed for Research in the Design of Secure Cyber Physical Systems. In: Proceedings of the 3rd International Workshop on Cyber-Physical Systems for Smart Water Networks, pp 25–28
- Anguita D, Ghio A, Oneto L, et al (2013) A public domain dataset for human activity recognition using smartphones. In: Esann, pp 3–4
- Anjum N, Latif Z, Chen H (2025) Security and Privacy of Industrial Big Data: Motivation, Opportunities, and Challenges. *Journal of Network and Computer Applications* 237:104130
- Ansari AF, Stella L, Turkmen C, et al (2024) Chronos: Learning the language of time series. arXiv preprint 240307815
- Bagnall A, Dau HA, Lines J, et al (2018) The UEA Multivariate Time Series Classification Archive, 2018. arXiv preprint 181100075
- Behrouz A, Santacatterina M, Zabih R (2024) Mambamixer: Efficient selective state space models with dual token and channel selection. arXiv preprint 240319888
- Benidis K, Rangapuram SS, Flunkert V, et al (2022) Deep learning for time series forecasting: Tutorial and literature survey. *ACM Computing Surveys* 55(6):1–36
- Bian Y, Ju X, Li J, et al (2024) Multi-patch prediction: Adapting LLMs for time series representation learning. arXiv preprint 240204852
- Bonifati A, Del Buono F, Guerra F, et al (2022) Time2feat: Learning interpretable representations for multivariate time series clustering. *Proceedings of the VLDB*

- Endowment (PVLDB) 16(2):193–201
- Brophy E, Wang Z, She Q, et al (2023) Generative adversarial networks in time series: A systematic literature review. *ACM Computing Surveys* 55(10):1–31
- Cai W, Wang K, Wu H, et al (2024a) Forecastgrapher: Redefining multivariate time series forecasting with graph neural networks. *arXiv preprint 240518036*
- Cai X, Zhu Y, Wang X, et al (2024b) Mambats: Improved selective state space models for long-term time series forecasting. *arXiv preprint 240516440*
- Centers for Disease Control and Prevention (2025) Weekly U.S. Influenza Surveillance Report. URL <https://www.cdc.gov/fluview/>
- Chen H, Li X, Liu W (2024) Multivariate Time Series Anomaly Detection by Fusion of Deep Convolution Residual Autoencoding Reconstruction Model and ConvLSTM Forecasting Model. *Computers & Security* 137:103581
- Chen L, Ng R (2004) On the marriage of lp-norms and edit distance. In: *Proceedings of the Thirtieth international conference on Very large data bases-Volume 30*, pp 792–803
- Cheng M, Liu Q, Liu Z, et al (2023) Timemae: Self-supervised representations of time series with decoupled masked autoencoders. *arXiv preprint 230300320*
- Cho K, van Merriënboer B, Gulcehre C, et al (2014) Learning phrase representations using RNN encoder-decoder for statistical machine translation. URL <https://arxiv.org/abs/1406.1078>, [arXiv:1406.1078](https://arxiv.org/abs/1406.1078)
- Choi J, Choi H, Hwang J, et al (2022) Graph neural controlled differential equations for traffic forecasting. In: *Proceedings of the AAAI conference on artificial intelligence*, pp 6367–6374
- Cini A, Marisca I, Zambon D, et al (2025) Graph deep learning for time series forecasting. *ACM Computing Surveys* 57(12):1–34
- Das A, Kong W, Sen R, et al (2024) A decoder-only foundation model for time-series forecasting. In: *Forty-first international conference on machine learning*
- Dau HA, Bagnall A, Kamgar K, et al (2019) The UCR Time Series Archive. *IEEE/CAA Journal of Automatica Sinica* 6(6):1293–1305
- Dempster A, Schmidt DF, Webb GI (2021) Minirocket: A very fast (almost) deterministic transform for time series classification. In: *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, pp 248–257
- Deza MM, Deza E (2009) Encyclopedia of distances. In: *Encyclopedia of distances*. Springer, p 1–583

- Dong J, Wu H, Zhang H, et al (2023) Simmtm: A simple pre-training framework for masked time-series modeling. *Advances in Neural Information Processing Systems* 36:29996–30025
- Du W, Côté D, Liu Y (2023) Saits: Self-attention-based imputation for time series. *Expert Systems with Applications* 219:119619
- Ekambaram V, Jati A, Dayama P, et al (2024) Tiny time mixers (ttms): Fast pre-trained models for enhanced zero/few-shot forecasting of multivariate time series. *Advances in Neural Information Processing Systems* 37:74147–74181
- Gama J, Žliobaitė I, Bifet A, et al (2014) A survey on concept drift adaptation. *ACM Computing Surveys (CSUR)* 46(4):1–37
- Gao J, Cao Q, Chen Y (2025) Auto-regressive moving diffusion models for time series forecasting. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, pp 16727–16735
- Gao S, Koker T, Queen O, et al (2024) Units: A unified multi-task time series model. *Advances in Neural Information Processing Systems* 37:140589–140631
- Garza A, Challu C, Mergenthaler-Canseco M (2023) Timegpt-1. *arXiv preprint* 231003589
- Goh J, Adepu S, Junejo KN, et al (2016) A dataset to support research in the design of secure water treatment systems. In: *International Conference on Critical Information Infrastructures Security*, pp 88–99
- Ha TW, Kim MH (2025) Multivariate time series anomaly detection using directed hypergraph neural networks. *Applied Artificial Intelligence* 39(1):2538519
- Hochreiter S, Schmidhuber J (1997) Long short-term memory. *Neural Computation* 9(8):1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Huang X, Chen W, Hu B, et al (2025) Graph mixture of experts and memory-augmented routers for multivariate time series anomaly detection. In: *Proceedings of the AAAI conference on artificial intelligence*, pp 17476–17484
- Hundman K, Constantinou V, Laporte C, et al (2018) Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding. In: *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data mining*, pp 387–395
- Ismail Fawaz H, Forestier G, Weber J, et al (2019) Deep learning for time series classification: a review. *Data Mining and Knowledge Discovery* 33(4):917–963
- Ismail Fawaz H, Lucas B, Forestier G, et al (2020) Inceptiontime: Finding alexnet for time series classification. *Data Mining and Knowledge Discovery* 34(6):1936–1962

- Jang Y, Byun J, Kwon S, et al (2024) K-raf: A Kubernetes-based resource augmentation framework for edge devices. In: Proceedings of the 33rd International Symposium on High-Performance Parallel and Distributed Computing, pp 364–366
- Jang Y, Kwon S, Choi I, et al (2025) A resource provisioning framework with adaptive task distribution for edge devices. In: NOMS 2025-2025 IEEE Network Operations and Management Symposium, pp 1–9
- Jia F, Wang K, Zheng Y, et al (2024) Gpt4mts: Prompt-based large language model for multimodal time-series forecasting. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp 23343–23351
- Jin M, Wang S, Ma L, et al (2023a) Time-llm: Time series forecasting by reprogramming large language models. arXiv preprint 231001728
- Jin M, Wen Q, Liang Y, et al (2023b) Large models for time series and spatio-temporal data: A survey and outlook. arXiv preprint 231010196
- Jin M, Koh HY, Wen Q, et al (2024) A survey on graph neural networks for time series: Forecasting, classification, imputation, and anomaly detection. IEEE Transactions on Pattern Analysis and Machine Intelligence 46(12):10466–10485
- Kolle, O., Kuebler, K., Huang, Y., et al (2021) Climate data (30min resolution) from the climate station in The Jena Experiment 2020. URL <https://www.bexis.uni-jena.de/JenaExperiment/Data/DataDetails/21042>
- Lai G, Chang WC, Yang Y, et al (2018) Modeling long- and short-term temporal patterns with deep neural networks. In: The 41st international ACM SIGIR Conference on Research & Development in Information Retrieval, pp 95–104
- Li Y, Chen W, Hu X, et al (2024) Transformer-modulated diffusion models for probabilistic multivariate time series forecasting. In: The Twelfth International Conference on Learning Representations
- Li Z, Rao Z, Pan L, et al (2023) Ti-mae: Self-supervised masked time series autoencoders. arXiv preprint 230108871
- Liang Z, Zhang J, Liang C, et al (2023) A shapelet-based framework for unsupervised multivariate time series representation learning. arXiv preprint 230518888
- Liang Z, Liang C, Liang Z, et al (2024) Timecsl: Unsupervised contrastive learning of general shapelets for explorable time series analysis. arXiv preprint 240405057
- Lim B, Arik SÖ, Loeff N, et al (2021) Temporal fusion transformers for interpretable multi-horizon time series forecasting. International Journal of Forecasting 37(4):1748–1764

- Lin J, Keogh E, Lonardi S, et al (2003) A symbolic representation of time series, with implications for streaming algorithms. In: Proceedings of the 8th ACM SIGMOD workshop on Research issues in data mining and knowledge discovery, pp 2–11
- Lines J, Taylor S, Bagnall A (2016) Hive-cote: The hierarchical vote collective of transformation-based ensembles for time series classification. In: 2016 IEEE 16th International Conference on Data Mining (ICDM), pp 1041–1046, <https://doi.org/10.1109/ICDM.2016.0133>
- Liu C, Xu Q, Miao H, et al (2024a) Timecma: Towards LLM-empowered time series forecasting via cross-modality alignment. arXiv preprint 240601638
- Liu C, Zhou S, Xu Q, et al (2025) Towards cross-modality modeling for time series analytics: A survey in the LLM era. arXiv preprint 250502583
- Liu J, Chen S (2024) Timesurl: Self-supervised contrastive learning for universal time series representation learning. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp 13918–13926
- Liu M, Liu Y, Liang K, et al (2023a) Deep temporal graph clustering. arXiv preprint 230510738
- Liu Q, Paparrizos J (2024) The elephant in the room: towards a reliable time-series anomaly detection benchmark. In: Proceedings of the 38th International Conference on Neural Information Processing Systems. Curran Associates Inc., Red Hook, NY, USA, NIPS '24
- Liu Q, Fang Y, Jiang P, et al (2024b) Dgcformer: Deep graph clustering transformer for multivariate time series forecasting. arXiv preprint 240508440
- Liu Y, Hu T, Zhang H, et al (2023b) itransformer: Inverted transformers are effective for time series forecasting. arXiv preprint 231006625
- Liu Z, Du B, Ye J, et al (2024c) An ncde-based framework for universal representation learning of time series. In: Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, pp 4623–4633
- Ma Z, López-Oriona Á, Ombao H, et al (2025) Fcpca: Fuzzy clustering of high-dimensional time series based on common principal component analysis. arXiv preprint 250507276
- Moges HT, Moodley D (2025) A lightweight spatial-temporal graph neural network for long-term time series forecasting. arXiv preprint 251217453
- National Renewable Energy Laboratory (2006) Solar power data for integration studies. URL <https://www.nrel.gov/grid/solar-power-data>

- Nie Y, Nguyen NH, Sinthong P, et al (2022) A time series is worth 64 words: Long-term forecasting with transformers. arXiv preprint 221114730
- Obata K, Kawabata K, Matsubara Y, et al (2024) Mining of switching sparse networks for missing value imputation in multivariate time series. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp 2296–2306
- Paparrizos J, Liu C, Elmore AJ, et al (2020) Debunking four long-standing misconceptions of time-series distance measures. In: Proceedings of the 2020 International Conference on Management of Data, SIGMOD Conference 2020, online conference [Portland, OR, USA], June 14–19, 2020. ACM, pp 1887–1905, URL <https://doi.org/10.1145/3318464.3389760>
- Paparrizos J, Boniol P, Palpanas T, et al (2022) Volume under the surface: a new accuracy evaluation measure for time-series anomaly detection. Proc VLDB Endow 15(11):2774–2787. <https://doi.org/10.14778/3551793.3551830>
- Paparrizos J, Yang F, Li H (2024) Bridging the gap: A decade review of time-series clustering methods. arXiv preprint 241220582
- Qiu X, Wu X, Lin Y, et al (2024) Duet: Dual clustering enhanced multivariate time series forecasting. arXiv preprint 241210859
- Rabiner LR (2002) A tutorial on hidden markov models and selected applications in speech recognition. Proceedings of the IEEE 77(2):257–286
- Reinsel D, Wu L, Gantz JF, et al (2019) The china datasphere: Primed to be the largest datasphere by 2025. International Data Cooperation (IDC), Framingham, MA, MA, USA, Rep US44613919 pp 1–10
- Runge J, Bathiany S, Bollt E, et al (2019) Inferring causation from time series in earth system sciences. Nature communications 10(1):2553
- Sakoe H, Chiba S (2003) Dynamic programming algorithm optimization for spoken word recognition. IEEE transactions on acoustics, speech, and signal processing 26(1):43–49
- Schäfer P, Höggqvist M (2012) Sfa: a symbolic fourier approximation and index for similarity search in high dimensional datasets. In: Proceedings of the 15th international conference on extending database technology, pp 516–527
- Shamba AK, Bach K, Taylor G (2025) Contrast all the time: Learning time series representation from temporal consistency. URL <https://arxiv.org/abs/2410.15416>, arXiv:2410.15416
- Shi F, Yin X, Wang K, et al (2025) Large language models for time series analysis: Techniques, applications, and challenges. arXiv preprint 250611040

- Soni J, Lange-Hegermann M, Windmann S (2025) Physics-informed diffusion models for unsupervised anomaly detection in multivariate time series. arXiv preprint 250811528
- Sørbo S, Ruocco M (2023) Navigating the metric maze: a taxonomy of evaluation metrics for anomaly detection in time series. *Data Min Knowl Discov* 38(3):1027–1068. <https://doi.org/10.1007/s10618-023-00988-8>
- Stefan A, Athitsos V, Das G (2012) The move-split-merge metric for time series. *IEEE transactions on Knowledge and Data Engineering* 25(6):1425–1438
- Su Y, Zhao Y, Niu C, et al (2019) Robust anomaly detection for multivariate time series through stochastic recurrent neural network. In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp 2828–2837
- Tashiro Y, Song J, Song Y, et al (2021) Csd: Conditional score-based diffusion models for probabilistic time series imputation. *Advances in neural information processing systems* 34:24804–24816
- Trindade A (2015) Electricityloadaddiagrams20112014. UCI Machine Learning Repository 10:C58C86
- Trirat P, Shin Y, Kang J, et al (2024) Universal time-series representation learning: A survey. arXiv preprint 240103717
- Um TT, Pfister FM, Pichler D, et al (2017) Data augmentation of wearable sensor data for parkinson’s disease monitoring using convolutional neural networks. In: *Proceedings of the 19th ACM International Conference on Multimodal Interaction*, pp 216–220
- Vlachos M, Kollios G, Gunopulos D (2002) Discovering similar multidimensional trajectories. In: *Proceedings 18th international conference on data engineering*, pp 673–684
- Wang F, Jiang Y, Zhang R, et al (2025) A survey of deep anomaly detection in multivariate time series: Taxonomy, applications, and directions. *Sensors* 25(1):190
- Wang J, Du W, Yang Y, et al (2024a) Deep learning for multivariate time series imputation: A survey. arXiv preprint 240204059
- Wang Y, Xu Y, Yang J, et al (2024b) Fully-connected spatial-temporal graph for multivariate time-series data. In: *Proceedings of the AAAI conference on artificial intelligence*, pp 15715–15724
- Wen H, Lin Y, Wu L, et al (2024) A Survey on Service Route and Time Prediction in Instant Delivery: Taxonomy, Progress, and Prospects. *IEEE Transactions on Knowledge and Data Engineering* 36(12):7516–7535

- Wen Q, Zhou T, Zhang C, et al (2022) Transformers in time series: A survey. arXiv preprint 220207125
- Wen Q, Chen W, Sun L, et al (2023) Onenet: Enhancing time series forecasting models under concept drift by online ensembling. *Advances in Neural Information Processing Systems* 36:69949–69980
- Woo G, Liu C, Sahoo D, et al (2022) Cost: Contrastive learning of disentangled seasonal-trend representations for time series forecasting. arXiv preprint 220201575
- Woo G, Liu C, Kumar A, et al (2024) Unified training of universal time series forecasting transformers. In: *Forty-first International Conference on Machine Learning*
- Wu Z, Pan S, Long G, et al (2019) Graph wavenet for deep spatial-temporal graph modeling. [arXiv:1906.00121](#)
- Wu Z, Pan S, Long G, et al (2020) Connecting the dots: Multivariate time series forecasting with graph neural networks. [arXiv:2005.11650](#)
- Xu H (2024) Hypergraph-based multi-scale spatio-temporal graph convolution network for time-series anomaly detection. arXiv preprint 241022256
- Xu J, Wu H, Wang J, et al (2021) Anomaly transformer: Time series anomaly detection with association discrepancy. arXiv preprint 211002642
- Yang X, Sun Y, Chen X, et al (2024) Frequency-aware generative models for multivariate time series imputation. *Advances in Neural Information Processing Systems* 37:52595–52623
- Ye L, Keogh E (2009) Time series shapelets: a new primitive for data mining. In: *Proceedings of the 15th ACM SIGKDD international conference on Knowledge discovery and data mining*, pp 947–956
- Ye Z, Chen M (2025) Diffusion Transformers for imputation: Statistical efficiency and uncertainty quantification. arXiv preprint 251002216
- Yeh CCM, Zhu Y, Ulanova L, et al (2016) Matrix profile i: all pairs similarity joins for time series: a unifying view that includes motifs, discords and shapelets. In: *2016 IEEE 16th international conference on data mining (ICDM)*, pp 1317–1322
- Yue Z, Wang Y, Duan J, et al (2022) Ts2vec: Towards universal representation of time series. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, pp 8980–8987
- Zamanzadeh Darban Z, Webb GI, Pan S, et al (2024) Deep learning for time series anomaly detection: A survey. *ACM Computing Surveys* 57(1):1–42

- Zeng A, Chen M, Zhang L, et al (2023) Are transformers effective for time series forecasting? In: Proceedings of the AAAI conference on artificial intelligence, pp 11121–11128
- Zeng C, Liu Z, Zheng G, et al (2024) Cmamba: channel correlation enhanced state space models for multivariate time series forecasting. arXiv preprint 240605316
- Zhang K, Wen Q, Zhang C, et al (2024a) Self-supervised learning for time series analysis: Taxonomy, progress, and prospects. IEEE Transactions on Pattern Analysis and Machine Intelligence
- Zhang X, Chowdhury RR, Gupta RK, et al (2024b) Large language models for time series: A survey. arXiv preprint 240201801
- Zhang Y, Yan J (2023) Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting. In: The Eleventh International Conference on Learning Representations, URL <https://openreview.net/forum?id=vSVLM2j9eie>
- Zhao L, Shen Y (2025) Proactive model adaptation against concept drift for online time series forecasting. In: Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1, pp 2020–2031
- Zheng W, Zhao P, Chen G, et al (2022) A Hybrid Spiking Neurons Embedded LSTM Network for Multivariate Time Series Learning under Concept-Drift Environment. IEEE Transactions on Knowledge and Data Engineering 35(7):6561–6574
- Zhou H, Zhang S, Peng J, et al (2021) Informer: Beyond efficient transformer for long sequence time-series forecasting. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp 11106–11115
- Zhou T, Ma Z, Wen Q, et al (2022) Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting. In: International Conference on Machine Learning, pp 27268–27286